The Hong Kong Polytechnic University
Department of Computing

COMP4913 Capstone Project
Report Final

Automatic Slide Publication for PowerPoint

Student Name:	Tse Hung Lap
Student ID No.:	24028633D
Programme-Stream Code:	61435-SYC
Supervisor:	Prof. PEI Yu MAX
Co-Examiner:	Dr LI Zecheng
2nd Assessor:	Prof. WANG Qixin
Submission Date:	10 April 2026

Abstract

The advent of synchronous hybrid pedagogy has highlighted key shortcomings in real-time content distribution and learner engagement. Current presentation tools lack automated publishing pipelines, requiring teachers to upload slides manually without real-time annotation or other interactive features needed for virtual classes. In this respect, this paper presents AutoSlide Publisher or simply AutoSlide made up of a PowerPoint VSTO Add-In, ASP.NET Core back-end, and AutoSlide Learning Management System (LMS) Web portal that together enable real-time slide publishing, per-slide unlocking, per-slide ink persistence, ink artifacts exportation, annotated PPTX pipelines, downloadable replay sessions, solution pages, and artificial intelligence-driven interactions.

Moreover, the AutoSlide platform features the Windows Runtime (WinRT) SpeechRecognizer component to enable real-time speech recognition, along with post-session on-demand summarization via the LLM Summarization Service and the LLM API.

The evaluation framework follows a Design-Based Research (DBR) approach and includes technical benchmarking and experimentation with 50 users in a validation environment; large-scale classroom use is currently outside the scope of this FYP project. Contribution to the field of hybrid-pedagogy research is a novel architectural model that integrates slide publishing, annotation capture, AI-enabled transcription and summarization, and replay within one native PowerPoint application.

Index Terms: Hybrid Education, PowerPoint Add-In, Real-Time Collaboration, AI Transcription, VSTO, Educational Technology

Contents

Abstract 2
List of Tables and Figures 4
List of Figures 4
List of Tables 4
1. Introduction 5
1.1 Background 5
1.2 Motivation 5
1.3 Problem Statement 6
1.4 Project Aim and Objectives 7
1.5 Project Scope 7
1.6 Project Timeline 9
2. Literature Review 10
2.1 Synchronous Hybrid Learning Environments 10
2.2 Presentation Automation Technologies 11
2.3 AI in Educational Interactivity 13
2.4 Office Add-In Development Frameworks 14
2.5 Research Gaps and Project Contribution 15
3. Methodology and System Design 17
3.1 Requirements 17
3.2 System Architecture 18
3.3 Runtime Behavioral Sequence {#runtime-behavioral-sequence} 20
3.4 Functional and Domain Modeling 23
3.5 Design Integration and Architectural Trade-offs 25
4. System Implementation and Realization 27
4.1 Implementation Status Overview 27
4.2 System Interface and User Workflows 27
4.3 Core Functional Modules Realized 30
4.4 Technical Challenges and Design Trade-offs 36
5. User Study & System Validation 37
5.1 Study Design and Participant Demographics 37
5.2 Task Design and Task-Completion Results 38
5.3 Quantitative Analysis: Usability and Perceived Value 38
5.4 Qualitative Feedback and Observer Notes 39
5.5 Threats to Validity 40
6. Future Work and Post-Project Refinements 41
6.1 Attendance Tracking Integration (Post-Validation Enhancement) 41
6.2 Collaborative Whiteboard and Split Teacher-Student Canvas 41
6.3 Vision-Based AI Explanation of Ink and Course Material 41
6.4 Multimodal Q&A Integration (Post-FYP Future Work) 42
7. Conclusion 43
References 44
Appendix A. User Study Questionnaire 46

List of Tables and Figures

List of Figures

FIGURE 1 PROJECT TIMELINE....9
FIGURE 2 SYSTEM CONTEXT DIAGRAM....19
FIGURE 3 DEPLOYMENT DIAGRAM....20
FIGURE 4 START LIVE SEQUENCE DIAGRAM....21
FIGURE 5 END SESSION SEQUENCE DIAGRAM....22
FIGURE 6 USE CASE DIAGRAM....23
FIGURE 7 DOMAIN MODEL DIAGRAM....24
FIGURE 8 SESSION STATE MACHINE DIAGRAM....25
FIGURE 9 CREATE NEW COURSE DIAGRAM....27
FIGURE 10 INSTRUCTOR MANAGE MATERIAL PAGE....28
FIGURE 11 ADD IN RIBBON - START PRESENTATION BUTTON....28
FIGURE 12 STUDENT DASHBOARD WITH LIVE SESSION....29
FIGURE 13 STUDENT LIVE VIEWER INTERFACE WITH UNDISCLOSED SLIDE LOCKED....29
FIGURE 14 STUDENT COURSE SESSION HISTORY FOR REPLAY AND DOWNLOAD MATERIAL....30
FIGURE 16 QR CODE AND JOIN SESSION MESSAGE BOX....32
FIGURE 17 INK CAPTURE OVERLAY WHERE INSTRUCTOR CAN DRAW SOLUTIONS....33
FIGURE 18 SAVED SOLUTION FOR STUDENT REVIEW....33
FIGURE 19 SIDE PANEL WITH LIVE TRANSCRIPTION....34
FIGURE 20 AI SUPPORTED TRANSCRIPT SUMMARY....35
FIGURE 21 STUDENT COURSE MATERIALS PAGE....35

List of Tables

TABLE 1: PROJECT OBJECTIVES....7
TABLE 2 PROJECT PHASES....10
TABLE 3 CONCEPT/FEATURE MATRIX OF REPRESENTATIVE EXISTING SYSTEMS....13
TABLE 4 FUNCTIONAL REQUIREMENTS AND TARGET DEFICIENCIES....18
TABLE 5 NON-FUNCTIONAL REQUIREMENTS....18
TABLE 6 SELECTED IMPLEMENTATION ANCHORS ACROSS THE AUTOSLIDE STACK....31
TABLE 7 TEST SUBJECT AND TEST BREAK DOWN....37
TABLE 8 TASK COMPLETION WITH HIGHEST SATISFACTION....38
TABLE 9 USER EXPERIENCE AND INTERPRETATION....39

1. Introduction

1.1 Background

Synchronous hybrid learning is a teaching method in which both in-class and remote students attend the lesson at the same time. Such an approach offers some advantages, although it requires proper design to achieve equal involvement and access [1].

On the other hand, the rise of the synchronous hybrid approach has revealed some issues inherent to presentation-based learning. PowerPoint is a widely used tool for lectures in higher education, and its use is embedded into a typical workflow of teaching [2]; however, traditional presentations lack the possibility to publish content live, view the material remotely in real time, and record the artifacts created during delivery.

As such, educational material is distributed after the lesson; therefore, late arrivals and those studying remotely cannot participate in the discussion or access class notes, which are essential for comprehending the content taught. The problem is exacerbated in hybrid classrooms, where remote participants are less engaged than their peers in class.

1.2 Motivation

The motivation for this project stems from three related observations regarding equity, pedagogical practices, and workflow productivity in synchronous hybrid classes.

The first observation is a mismatch between live teaching sessions and the publication of those lectures. When an instructor teaches in real time, he or she adds ink, highlighting, and sketches to convey aspects of the topic that cannot be conveyed entirely on the slide itself. If the slides are published after the class ends, remote students will receive an abridged version of the lecture, while on-campus students will receive the enriched experience. This can harm equitable learning experiences.

The second issue is that the current presentation-sharing tools do not offer a way to progressively disclose slides. In our context, we define progressive disclosure as a method in which a teacher has complete control over the process of showing slides, and students only see what has already been taught. A staged information release could help optimise the teaching process by managing attention and preventing unnecessary processing of information [3], [1]. However, typical commercial solutions allow access to full presentations, which may undermine carefully designed lesson progression.

Finally, interactive capabilities still have to be used separately. Transcriptions, polls, questions and answers, and artificial intelligence summarizing capabilities are typically provided by third-party providers. Fragmented toolchains lead to additional extraneous load, since the learner has to constantly pay attention to several different information sources and their interfaces [3].

Therefore, the problem here is not only a lack of features but also integration issues. Instructors will need a workflow that stays within PowerPoint, while students will need a synchronized presentation.

1.3 Problem Statement

Current presentation delivery in hybrid classrooms suffers from three critical deficiencies that this project aims to address:

Deficiency 1: Manual publication overhead and loss of live teaching artifacts.

It is common for teachers to required to manually post lecture material after the lectures. This leads to unnecessary work for both students and teachers and denies students access to lecture material during real-time delivery. However, more significantly, this process does not capture live-annotated material, which is critical pedagogical information, especially in STEM subjects where visual presentations, mark-ups, and explanations are essential [4], [5].

Deficiency 2: Lack of secure progressive disclosure.

Existing slide-sharing tools do not provide presenter-controlled locking and unlocking of slides. 3As a result, unrevealed material may be exposed prematurely, which conflicts with instructional strategies that depend on staged information release and scaffolded progression [3], [1].

Deficiency 3: Disconnected interactions and artifact creation.

Features like transcription, summarization, polling, and question answering are usually offered separately from the teaching process. . This fragmentation forces instructors to switch contexts repeatedly, increases complexity in the classroom, and hinders the creation of cohesive post-classroom artifacts like annotated slides, transcriptions, and summaries. Disconnected toolkits may even introduce additional extraneous cognitive load, splitting students' focus across different information streams and user interfaces [3].

These deficiencies collectively reduce remote student engagement, exacerbate accessibility challenges for students requiring assistive technologies or cultural/linguistic accommodation, and impose unsustainable workload on educators managing multiple disconnected systems.

1.4 Project Aim and Objectives

Project Aim: Develop AutoSlide, a PowerPoint VSTO add-in and associated platform that supports automatic real-time slide publication, secure progressive unlocking, controlled access, and AI-enhanced interactivity within a PowerPoint-native hybrid teaching workflow.

This project contributes

(i) a PowerPoint-native workflow for real-time publication with per-slide access control and artifact capture,
(ii) an end-to-end VSTO--ASP.NET Core–web architecture pattern for hybrid teaching delivery and replay, and
(iii) implementation and evaluation evidence that informs future refinement of AI-assisted classroom publishing systems.

Project Objectives:

Objective	Description
O1	Real-time automated publication: Develop slide synchronization with less than 2 s latency (95th percentile)
O2	Secure progressive unlocking: Design server-side access control mechanisms for enforcing presenter-controlled slide progression
O3	Live annotation capture: Develop ink detection and export pipeline leveraging PowerPoint COM API
O4	AI-powered transcription: Integrate Windows Runtime (WinRT) SpeechRecognizer for local privacy-preserving speech-to-text
O5	Automatic summarization: Generate summarizations based on the LLM Summarization Service with external LLM API
O6	Post-session artifacts: Deliver PPTX slides with ink strokes, transcript, PDFs, and a standard data schema for exporting Q&A results from other sessions in the future.

Table 1: Project Objectives

1.5 Project Scope

Inclusions Scope:

- PowerPoint VSTO Add-In providing functionality for session management, slide export, ink capture, and presentation control

- Backend Service handling real-time synchronization with SignalR, session management, storage management, and AI capability

- Web Viewer application allowing students to interact with the synced slides and other functionality
- LLM Summarization Service with external API integration
- Ink capture, export, playback, and Progressive Disclosure Workflow
- Design-Based Research (DBR) as per the technical benchmarking and validation process
- Privacy-by-design approach with speech processing on device, controlled access, and minimum data retention policies

Exclusions Scope:

The following features have been deliberately excluded from the FYP scope and will be handled during post-FYP stages of development and research:

- Classroom deployment with N>100 participants – research post-FYP
- Multimodal Q&A interaction that involves equation/shape recognition by AI - Post-FYP Future Work
- Database operation hardening including backups and high availability - operations post-FYP
- Background Job Scheduler (Hangfire) to enable roll summary generation - Future Enhancement Feature
- Cross-platform office add-in with JavaScript API version for MacOS/WEB - alternative future architecture
- Integration with institution SSO and deployment procedure - institutions phase post-FYP
- Native mobile applications on iOS/Android - Out-of-scope (viewer app is responsive)
- Complex drawing recognition involving computer vision and machine learning techniques – Post-FYP Future Work
- Scalability assessment and testing on large user samples (N>100) – post-FYP research

1.6 Project Timeline

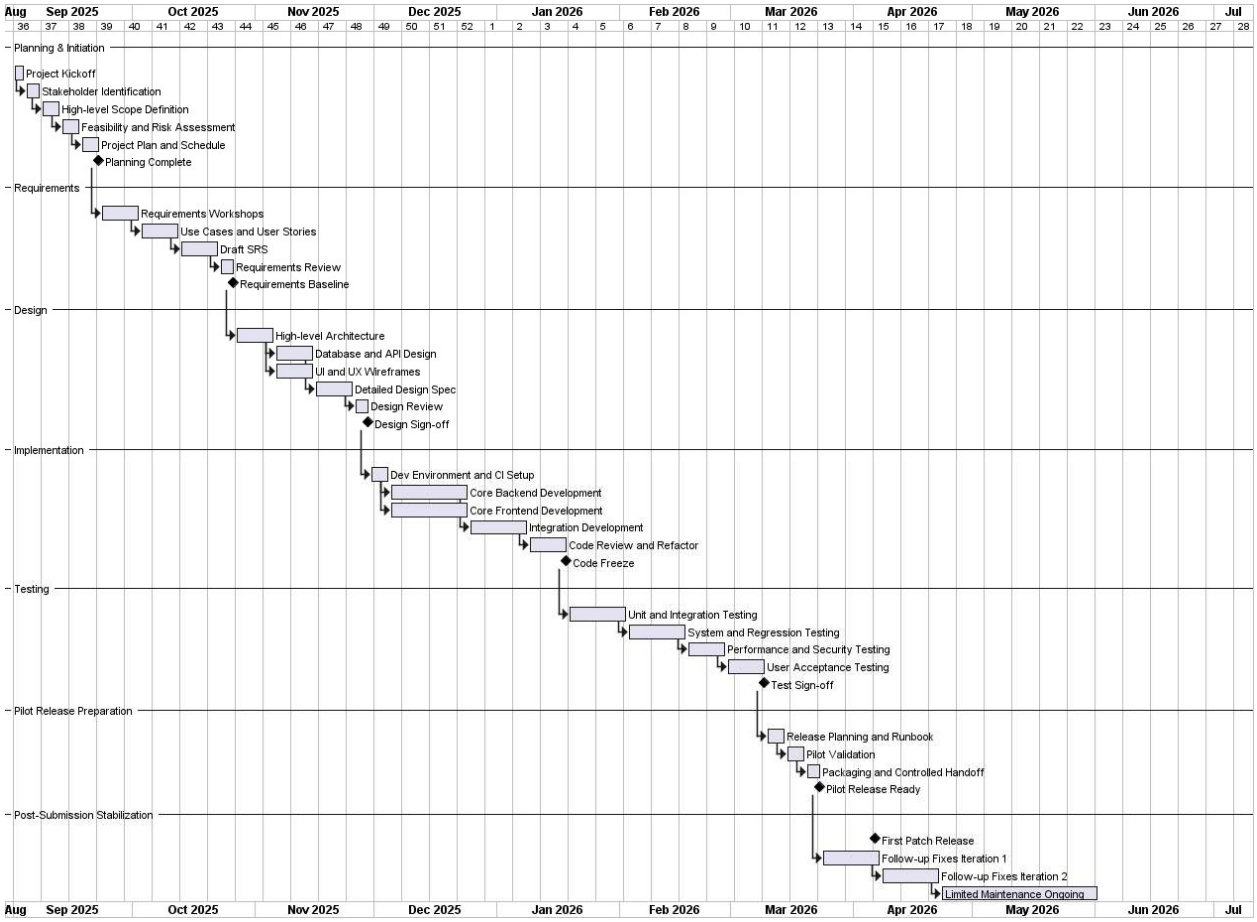

Figure 1 Project Timeline

The duration for which the project is scheduled through the use of the Gantt chart is from September 1st, 2025, till July 10th, 2026. The Gantt chart has planned for the task such that planning precedes requirements, requirements precede design, design precedes implementation, and code freeze precedes testing and pilot-release preparation.

Phase	Planned window	Main activities	Engineering outcome
Planning & Initiation	Sep 2025	kickoff, stakeholder identification, scope definition, feasibility and risk assessment, schedule setting	project baseline and delivery plan
Requirements	late Sep–Oct 2025	requirements workshops, use cases and user stories, draft SRS, review, baseline approval	frozen functional scope
Design	late Oct–Dec 2025	system architecture, database and API design, UI wireframes, detailed design, design review	implementation-ready architecture
Implementation	Dec 2025–Feb 2026	development environment and CI setup, backend development, frontend development, integration, refactoring, code freeze	integrated prototype

Testing	Feb–Apr 2026	unit and integration testing, system and regression testing, performance and security testing, user acceptance testing	validated release candidate
Pilot Release Preparation	Apr 2026	release planning, pilot validation, packaging, controlled handoff	pilot release readiness
Post-Submission Stabilization	May–Jul 2026	follow-up fixes, limited enhancement iterations, ongoing maintenance	post-submission stabilization

Table 2 Project Phases

The academic submission window closes in May 2026, but the engineering timeline extends into July 2026 to accommodate follow-up fixes and limited enhancement work after submission. Large-scale classroom deployment remains outside the validated FYP scope, and the schedule therefore retains only a short post-submission stabilization tail for operational follow-up.

2. Literature Review

This literature review serves as the foundation of the theory and technology behind AutoSlide. It begins by discussing the issues of synchronous hybrid learning and engagement imbalance (Section 2.1). Then, it provides an overview of presentation automation technologies and live-viewing software (Section 2.2). This is followed by the discussion on AI-based interactivity, including transcription and summarization (Section 2.3). Finally, it discusses the Office add-ins technology, particularly those relevant to PowerPoint-based implementations (Section 2.4).

2.1 Synchronous Hybrid Learning Environments

Synchronous hybrid learning—simultaneous instruction of co-located and remote students has become an increasingly important mode of higher-education delivery. Raes et al. (2020) conducted a systematic literature review analyzing 37 empirical studies and identified key benefits including flexibility, accessibility, and support for inclusive pedagogy.[1] At the same time, the review highlighted persistent challenges, including interaction asymmetry between on-site and remote cohorts, technological complexity that increases instructional burden, and weakened social presence for online participants.

Moreover, Palmer et al. (2022) further demonstrate that the asymmetry is not a hypothetical concept. For instance, in their mixed-methods study on synchronous hybrid instruction in pharmacy education, remote learners engaged in active listening and emotional reactions less than those attending in person.[6] However, the lack of significant differences in note-taking, question-asking, and group work indicates that the participation discrepancy is not fixed and can be alleviated through the implementation of instructional and technological strategies.

Similar conclusions can be drawn from the research findings of Aiken et al. (2025), who argue that blended and hybrid learning can be effective when learners and instructors focus on social interaction, equitable access, and effective learning design.[7] Overall, all these works show that synchronous hybrid learning is both feasible and advantageous for education but requires

supportive technology to preserve its instructional timing, equalize participation opportunities, and narrow the gap between in-class and remote learners.

To date, no literature reports an integrated approach to creating a system that would combine live publication of slides, live artifact storage, and interaction assistance powered by artificial intelligence in the same interface. Such limitation has an obvious connection to Deficiencies 1 and 3 mentioned in Section 1 since AutoSlide must provide artifact availability and participation tools to alleviate remote participation asymmetry.

2.2 Presentation Automation Technologies

Research on presentation automation has largely evolved along three separate streams: AI-assisted slide authoring, classroom response or audience response systems (ARS), and web-based slide delivery or replay tools. These streams improve isolated parts of the presentation lifecycle, but they do not yet constitute a unified live-teaching pipeline.

Chen et al. (2023) evaluated AI-assisted slide generation systems and reported a 37% reduction in preparation time.[8] However, these benefits are concentrated in pre-presentation authoring rather than live delivery. In other words, the tools help users create slides faster, but they do not address synchronized publication, annotation persistence, or session-linked artifact generation during teaching.

Thus, it seems there is a structural separation between the toolsets used for pre-class preparation and those used for in-class delivery. Most tools have focused on either optimizing the creation process or assisting with slide display, but have not considered the entire life cycle, starting from slide advancement and leading up to the real-time publishing, annotation capturing, slide replay, and artifact export through a single PowerPoint plugin. This suggests that AutoSlide should treat slide advancement as a publishable event and retain live artifacts as primary outputs.

2.2.1 Existing Product Landscape and Feature Coverage

To complement the academic literature on the automation of presentations, it would be helpful to contrast some sample existing systems to the function pipeline considered in this work. Previous research indicates that synchronous hybrid teaching can be successful only where access, equity, and seamless engagement between classroom and off-site participants is provided by tools used for it [1], [6], [7]. At the same time, the modern landscape consists of disparate approaches that have yet to merge around PowerPoint functionality [8], [9], [10].

System / category	Real-time slide sharing	Progressive disclosure	Live ink / whiteboard capture	Persistent replay / artifact continuity	Integrated AI support	PowerPoint-native workflow	Main limitation relative to AutoSlide
Microsoft Teams + PowerPoint Live [11]	✓	✗	△	✗	△	✓	Strong for slide delivery and accessibility, but not a unified pipeline for secure progressive release, persistent ink artifact export, and integrated post-session outputs.
Zoom Whiteboard [12]	✗	✗	✓	✓	✓	✗	Strong for collaborative canvases and persistence, but not for PowerPoint-native slide control or synchronized slide publication.
AI-assisted slide authoring tools [8]	✗	✗	✗	✗	✓	✗	Improve pre-class preparation, but do not support live delivery, annotation preservation, or session-linked artifact generation.

Automatic captioning / summarization tools [9], [10]	✗	✗	✗	△	✓	✗	Improve accessibility and review, but are typically external to the live presentation workflow.

Table 3 Concept/feature matrix of representative existing systems.

The structural gap is clear in Table X. Current tools provide functionality to some degree along the pipeline, but not to others. For instance, PowerPoint Live focuses on live broadcasting and accessibility [11], Zoom Whiteboard focuses on persistent visual collaboration [12], while tools oriented toward artificial intelligence cater to either the creation process or post-processing phases [8], [9], [10]. None of the above tools, according to the current literature and vendor documents examined in this project, supports an end-to-end PowerPoint-native workflow consisting of live slide publishing, presenter-controlled unveiling, ink persistence, recording, and AI-powered interactivity.

2.3 AI in Educational Interactivity

Interactivity supported by artificial intelligence technologies is currently at the advanced stage of development, primarily in three fields pertinent to this research topic: transcribing and captioning for accessibility purposes, summarization, and innovative multimodal analysis of visual artifacts. While there is evidence in favour of all three aspects in literature, they are typically considered independently rather than integrated into a single workflow for delivering lectures.

Classroom transcription and captioning have received considerable attention as an accessibility measure for students with disabilities (deafness or hearing impairments) and for students with language barriers (foreign students). In a randomised controlled experiment involving 62 deaf adults in an online course, Yoon and Kim found that captions helped students better understand lecture material, compared with controls without captions, although cognitive load and motivation did not differ between groups [9]. Research on classroom live captioning indicates that text equivalents greatly enhance the access of deaf and hard-of-hearing students to educational materials and benefit their hearing counterparts in poor acoustic environments. At the same time, such systems should take into account the need to maintain transcription accuracy and the ability to correct errors promptly, in order to be practically applicable in the instructional process.

The relevance of this field to the study lies in Alhusban's research, which examined a rotational hybrid approach to classroom participation that alternates between face-to-face and virtual

attendance in the same session. Although most participants had a positive attitude towards it, they perceived the learning opportunities as similar across modalities.[13]

Automated summarization has also been studied as a way to reduce information overload while retaining key concepts. Shimada et al. report that, in a study involving more than 300 university students, those who previewed summarized slide sets achieved higher pre-lecture quiz scores than peers who previewed the full slide deck, despite spending less time preparing.[10] At the same time, research on analysis paralysis warns that excessive or poorly structured information can impair decision-making and comprehension, which means that summarization must be selective, timely, and presentation-aware rather than indiscriminate.[14]

A further emerging strand concerns multimodal AI. Lee and Zhai demonstrate that GPT-4V can answer questions about educational visuals, including student-drawn scientific diagrams and classroom settings, without requiring a custom-designed computer vision pipeline.[15] For this document, multimodal AI is more relevant as an additional future research idea beyond the core FYP focus than as a capability that needs to be supported in this report.

In general, current literature supports the usefulness of individual features like transcription, summarization, and multimodality, but fails to show how to coordinate those in a live presentation workflow under constraints of latency, usability, and correction. Hence, for this report, AutoSlide focuses on the integration of transcription and summarization into the presentation slide flow to avoid tool switch and leaves multimodal AI reasoning as future research work.

2.4 Office Add-In Development Frameworks

The extensibility of Microsoft Office within the current research effort is based on two primary approaches: Visual Studio Tools for Office (VSTO) and the Office JavaScript API. The selection of any approach is not a question of personal preference but rather affects the type of events to capture, object access, and other aspects of runtime control.

Specifically, using VSTO (.NET Framework), a developer can access the managed COM API, allowing for fine-grained PowerPoint events such as Application.SlideShowNextSlide or accessing properties associated with ink in specific slides.[16] As a result, VSTO allows capturing more events and exercising greater runtime control over PowerPoint application components. However, there are certain drawbacks to VSTO implementation – for example, platform limitations (Windows only) and dependencies on .NET Framework components (which may require manual installation).

In contrast, using the Office JavaScript API enables cross-platform deployment on Windows, macOS, and even web platforms, combined with centralized distribution and packaging of add-ins.[17] However, Office.EventType.DocumentSelectionChanged is not as precise when

capturing live slide-show events. Additionally, extracting data associated with slide ink requires asynchronous methods rather than directly accessing the PowerPoint COM API.

According to Henriksen et al. (2023), hybrid architectures based on Office-side integration and backend services and viewers written in ASP.NET are feasible in real-world applications. Such findings suggest that native Office integration does not preclude the deployment of add-ins in browsers or on other platforms.

For applications that require slide-show events, synchronization access, and tight integration with Office's object models, VSTO appears to be the most viable option from those documented in the literature. However, VSTO, ASP.NET Core, SignalR, and browser viewers have not yet been integrated into a coherent educational architecture in the extant literature. Consequently, AutoSlide uses VSTO to capture instructor and delegate events, and delegates other aspects of the workflow to backend and viewer components.

2.5 Research Gaps and Project Contribution

Synthesizing the literature reveals four critical gaps:

Gap 1: Integrated Automation

There is no integrated workflow for publishing slides, capturing annotations, replaying annotations, and facilitating AI interaction in PowerPoint's native teaching environment. The literature considers authoring tools, audience participation platforms, and slide web viewers separately, not collectively.

Gap 2: Progressive Disclosure and Access Control Models

Current viewers tend to display a slide deck's entire content once it's shared, without allowing the presenter to control the progression. This makes the idea of staged instruction and progressive content access unsupported for hybrid teaching.

Gap 3: Ink Capture in Slide Show Mode

While static ink is well supported by PowerPoint's various workstreams (including education and office environments), dynamic (live) ink capture in active slideshow mode is technically challenging because overlay rendering and committed slides do not naturally synchronize.

Gap 4: Workflow-Level AI Orchestration

The literature discusses transcript and summary capabilities as useful additions to a teaching experience, but does not provide guidelines for integrating them seamlessly without introducing significant delay, correction effort, or interface fragmentation.

Gap 5: VSTO-Web Architecture Patterns

There are a few examples of best practices for using PowerPoint VSTO add-ins with a backend service and a browser viewer in a reliable, real-time environment.

This project aims to address these gaps through a practical engineering contribution. AutoSlide addresses the identified gaps by providing: (1) an architecture of a PowerPoint VSTO add-in that can enable a seamless live slide publication and annotation export workflow, (2) the server-side mechanisms that ensure progressive unlocking of slides, (3) the workflow for capturing, persisting, and exporting annotations live within the slide-show mode, (4) the workflow for collecting and reviewing transcripts and summaries, and (5) a replicable pattern for developing educational tools with a VSTO-ASP.NET Core-React architecture.

To be more specific, each of the gaps listed above can be used as a justification for a particular set of functional requirements identified in Section 3:

- Gap 1 can justify FR-SESSION, FR-UNLOCK, FR-INK, and FR-REPLAY as the system will require real-time slide publication, annotation capture, and replay functionalities in a unified workflow.
- Gap 2 can justify FR-UNLOCK and FR-AUTH as the viewer will need to implement the per-slide access control as well as session-level authentication functionality.
- Gap 3 can justify FR-INK and FR-REPLAY as the live ink artifacts should be captured and stored in a session while being replayable later.
- Gap 4 can justify FR-AI and NFR-SEC as the system will require transcription and summarization support, making the output accessible for review while respecting users' privacy.
- Gap 5 can justify FR-SESSION, FR-REPLAY, FR-AI, and NFR-REL as the architecture must support a connection between the PowerPoint add-in, the backend service, and the browser-based viewer.

The identified gaps above will become the main constraints for the design of AutoSlide: PowerPoint-native events, server-based session management, per-slide access enforcement, persistent annotations, and AI integration. These constraints are further detailed in Section 3.

3. Methodology and System Design

This chapter translates the deficiencies identified in Section 1 and the literature gaps synthesized in Section 2 into an implementable system design. In particular, the AutoSlide system is formally defined using three distinct perspectives: (i) architectural drawings which outline system boundaries and trust regions, (ii) run-time scenarios that confirm temporal validity in the presence of asynchronous events, and (iii) use case/domain models that allow for replay and late-joining.

Methodology of Development: The software development process adopted in this project was iterative and incremental across FYP Part A and FYP Part B. In the initial phase, requirements analysis and architecture were done, and in the later phase, there were multiple cycles of design, coding, testing, and fixing. Iterations were based on monthly reviews with the project supervisor, who provided educational and practical feedback on how to improve the system architecture accordingly. At each iteration, requirements, diagram sets, and a new demo were produced. The iterations were initially based on the supervisor's feedback period and later complemented by the systematic testing framework described in Section 5.

3.1 Requirements

3.1.1 Functional Requirements

ID	Requirement	Deficiency addressed
FR-COURSE	Course and session management: Instructors create and maintain courses and rosters; Students have access only to courses they have joined and sessions that are currently live.	D3
FR-SESSION	Secure real-time slide delivery: Instructors upload PPTX files, they get converted by the backend asynchronously; Slide indices get published by SignalR; Joining is available only for enrolled users; Presenters can explicitly control starting, stopping, or terminating slides publishing, so there is no visibility without their explicit action.	D1
FR-UNLOCK	Slide unlocking and navigation control: The system manages unlocking status per slide, prohibits navigation to non-unlocked slides, and maintains session state across browser refresh and late joining.	D2
FR-INK	Ink capture, synchronization, and storage: Add-in renders the ink layer, associates strokes with slides, stores ink data per slide, restores it on returning to a slide, and cleans up on session termination.	D1
FR-REPLAY	Web viewing and replay: Students see the synchronized real-time view; After that they see the timeline of slides, their transcript, and authorized materials download link.	D1

FR-AUTH	Authentication and authorization: JWT authentication, access roles for dashboards and artifacts, change password functionality.	D2
FR-AI	Transcription and summarization of sessions: On-device speech recognition, which sends text fragments to the server asynchronously; Transcripts get saved and displayed; Summary of each session based on LLM is provided.	D3
FR-MATS	Materials management: Course materials upload by instructors and distribution; Downloadable course material for students; Solution page creation by session link.	D1

Table 4 Functional Requirements and target deficiencies

3.1.2 Non-functional Requirements

The non-functional requirements express the quality attributes that constrained the architecture, data model, and UI design in Part 3.

ID	Requirement
NFR-PERF	Performance: live slide changes should remain near real time, and presentation upload/conversion should not block the presenter workflow.
NFR-REL	Reliability: slide index updates must be idempotent, sessions must survive reconnects cleanly, and AI/upload failures must not crash the presenter view. For AI-related failures (transcription or summarization), the system must degrade gracefully by preserving core slide and ink delivery even when transcripts or summaries are temporarily unavailable.
NFR-SEC	Security and privacy: no local path leakage in URLs or logs, enrollment-based access control must protect sessions and artifacts, and transcripts should follow privacy-by-design principles.
NFR-USAB	Usability and accessibility: the web viewer should be understandable for non-technical users, and live/replay states should be visually distinct.
NFR-COMP	Compatibility: the instructor side must run on Windows via VSTO, while the student side should work in modern browsers.
NFR-MAINT	Maintainability and extensibility: backend features, frontend pages, and add-in logic should remain modular and documented for future contributors.

Table 5 Non-functional Requirements

3.2 System Architecture

AutoSlide follows the desktop-cloud-web three-layer architectural approach, where the instructor interacts within the PowerPoint environment, whereas the AutoSlide backend and web portal constitute the environment of persistence, authorisation, and playback functionality. In this approach, the live presentation critical path remains minimal while keeping the scope open for describing the system in terms of its context and implementation.

3.2.1 System Context

FYP/out/docs/diagram/system_context_diagram/system_context_diagram.png

In the context of the system, the AutoSlide architecture comprises the following five components: PowerPoint VSTO Add-In, AutoSlide Backend, AutoSlide Web Portal, Supabase (Postgres + Object Storage) and External LLM API. The instructor interacts with AutoSlide via PowerPoint, whereas the student does so via the web browser, and there may also exist an external LMS/ADMIN API Service which provisions courses and rosters externally.

With the backend separating the two client types, this enables HTTPS communication, SignalR, storage, and API calls, avoiding clutter on the instructor's computer. It makes the development testable, as one can mock or stub external dependencies without considering how the PowerPoint Add-In workflow or the Student Web client interface would interact with any particular dependency.

3.2.2 Component Architecture

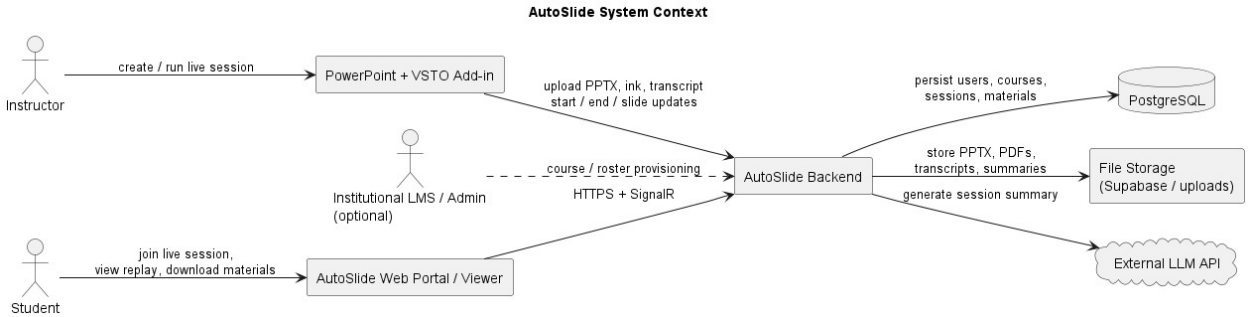

Figure 2 System Context Diagram

Further breaking down the presentation add-in into SessionManager, InkOverlayService, SpeechService, PresentationApiGateway, and SignalRService, as well as the backend into UploadService, LiveSessionController, SessionsController, PresentationHub, PresentationStore, SessionSummaryWorker, OpenRouterChatService, and AppDbContext and finally the web portal into LiveSessionViewer, StudentCourseView, and ReplayViewer results in separation of concerns.

For instance, PresentationHub performs fanout, AppDbContext performs EF Core relational persistence, and SessionSummaryWorker performs the expensive LLM call path separate from slide transition and authentication operations. With such an architecture, testability becomes easier for modules that are decoupled from each other, and changes to storage and AI services can be made independently of real-time session processing logic.

3.2.3 Deployment Architecture

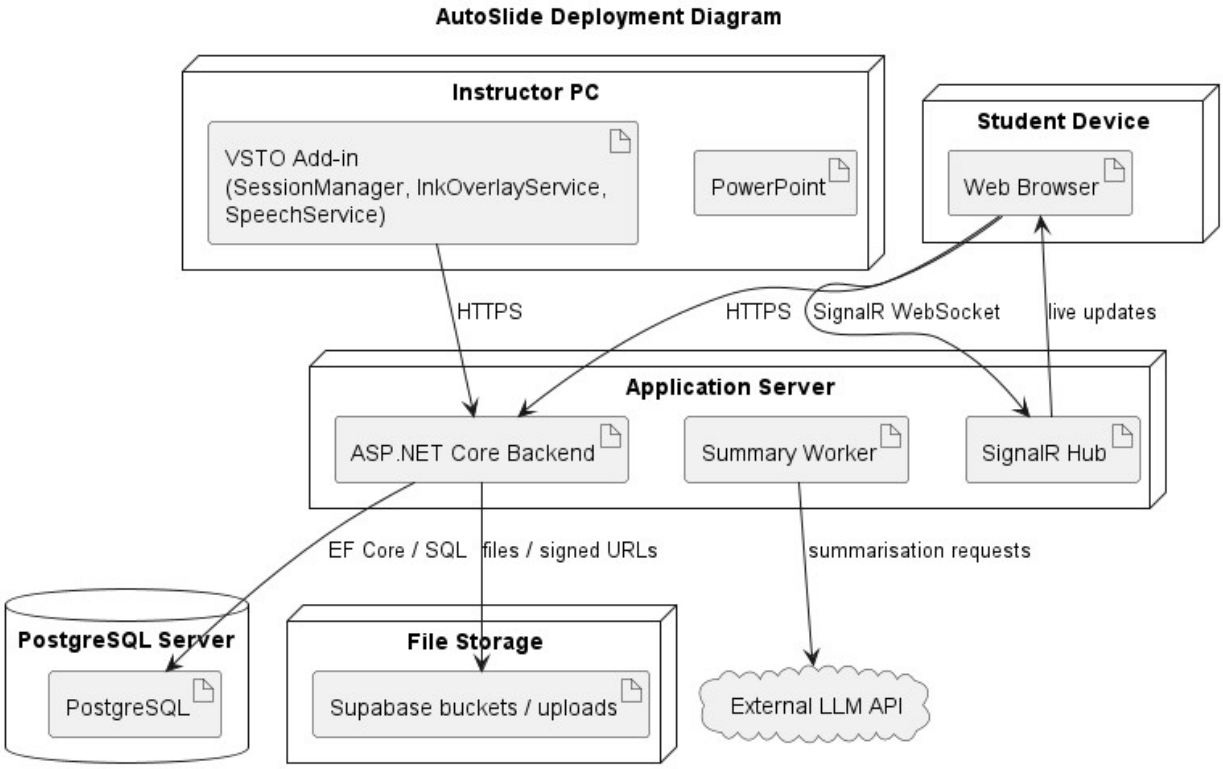

Figure 3 Deployment Diagram

Finally, the deployment diagram lists all runtime host environments: instructor PC, student device, application server, Supabase (PostgreSQL + object storage), and an external LLM API. PowerPoint and the VSTO add-in both run exclusively on the instructor PC, while the browser runs exclusively on the student device, and the ASP.NET Core backend, SignalR Hub, and SessionSummaryWorker run on the Application Server.

Supabase has been chosen as the underlying infrastructure for relational persistence and object storage, owing to its managed PostgreSQL support and built-in object storage for PDF transcripts, annotated PPTX exports, and replay assets. It enables a clear boundary to be set around the backend, which can communicate only via backend services with the add-in and the browser viewer. Meanwhile, EF Core persists relational data, and the server manages storage writes. In contrast, implementing and managing database and object storage services separately added considerable friction to development, implementation and testing.

In this configuration, however, there is still a trade-off between PowerPoint integrations and cross-platform compatibility, since the instructor interface must remain Windows-exclusive while the student side remains browser-based. On the other hand, this separation helps with fault isolation and scalability for the backend independent of the rest of the components. The downside of such a choice is that additional operational hardening of the managed storage layer, including backup strategies, high availability, quota management, and tuning for production scaling, falls out of scope for the current FYP.

3.3 Runtime Behavioral Sequence {#runtime-behavioral-sequence}

The presented diagrams demonstrate the process of moving from session initialization through live synchronization up to archiving the ready-to-playback version. The diagrams are critical because they clearly demonstrate that the architecture supports asynchronous file uploads, SignalR fan-out, and asynchronous post-session artificial intelligence-based processing of transcript without blocking the presenter's workflow.

3.3.1 Start Session

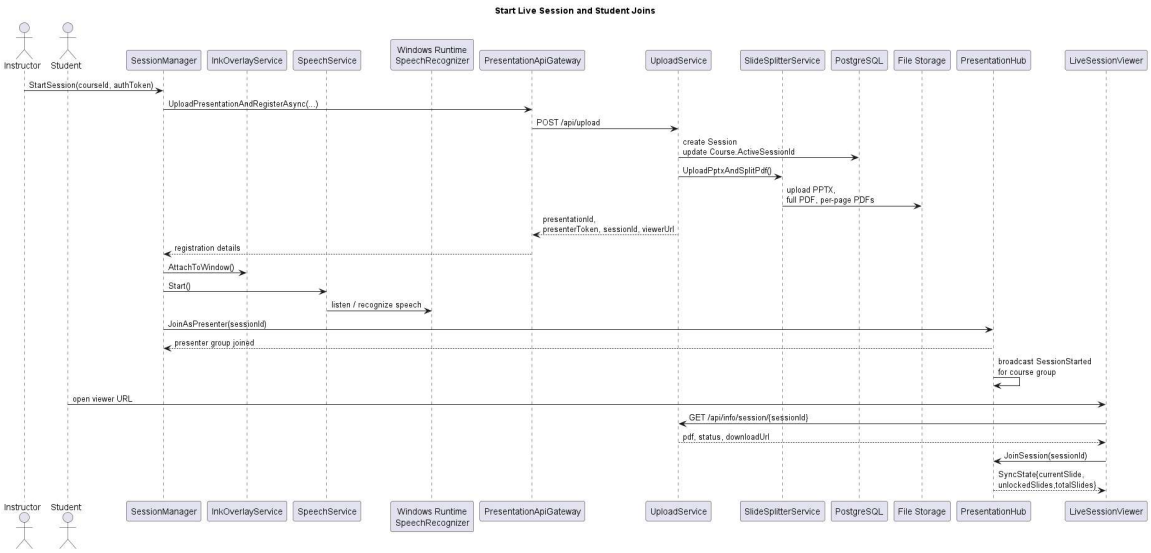

Figure 4 Start Live Sequence Diagram

In the first step, SessionManager invokes PresentationApiGateway in order to upload the presentation and create a session. The second step involves uploading the session via the UploadService by writing the row and updating the Course.ActiveSessionId and sending PPTX to the SlideSplitterService in order to create PDF slides. In turn, the add-in attaches the ink overlay, starts SpeechService, and connects to the presenter group within PresentationHub while the student viewer requests session data and joins the SignalR session.

It should be noted that the described flow enables asynchronous file management; as a result, the add-in will not have to wait for PDF splitting to occur before enabling the live session. On the other hand, the backend will have to ensure synchronization on its side; therefore, it will be easier to avoid blocking the UI and to allow latecomers to join the session from the same slide. In addition, the live session can be started earlier than the PDF splitting delay.

3.3.2 Live Navigation

When the presenter shows a new slide or draws a line with ink, SessionManager informs LiveSessionController about the slide or unlock, and PostgreSQL writes the indexes CurrentSlideIndex and SessionUnlockedSlide, after which PresentationHub sends SlideUnlocked to the client application. At the same time, InkOverlayService transmits the ink drawing data via SignalRService, and SpeechRecognizer sends the recognised speech transcript to SpeechService, from where it reaches the server-side via POST /api/transcript.

Thus, the live loop is free from having to invoke high-frequency events on the polling path, as they can stay on the SignalR path, thereby preventing UI jitter. It is possible to maintain FR-UNLOCK, FR-INK, and FR-AI, and to offer the ability to trigger the server-side path multiple times per slide event. If transcription is interrupted or fails at any point, or if transcript fragments are missing, the live slide and ink channels can continue functioning properly, enabling lectures to proceed without the auxiliary AI text recognition.

3.3.3 End Session and Replay

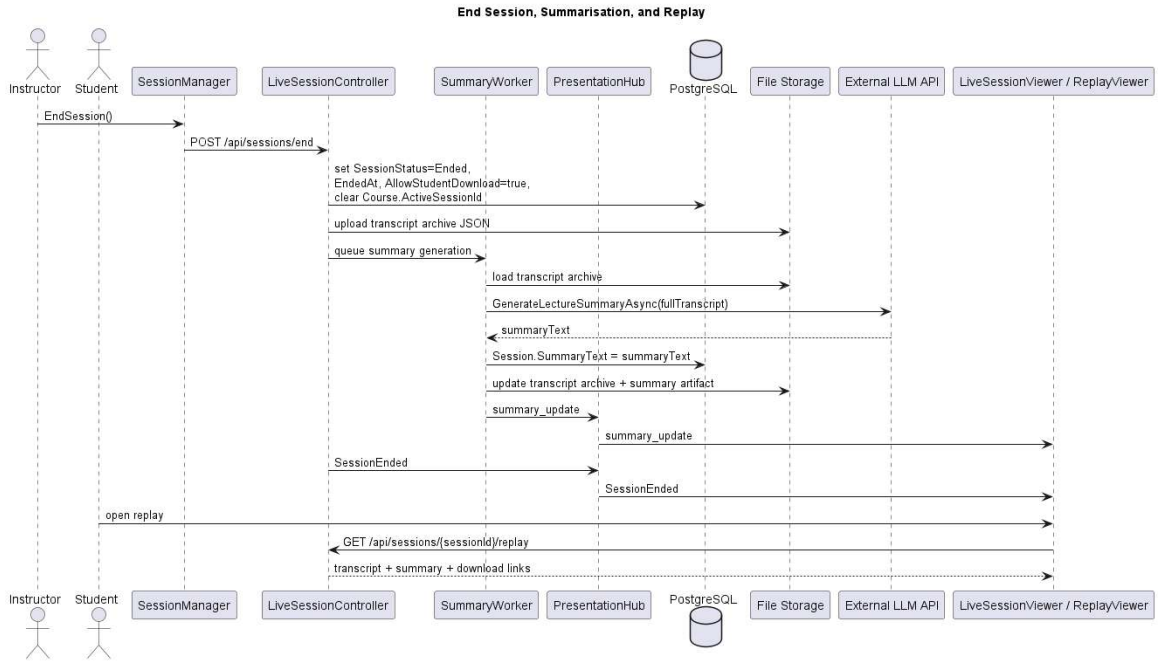

Figure 5 End Session Sequence Diagram

In contrast, the end session path starts by invoking LiveSessionController to terminate the current session, changing its status to Ended, adding the ended timestamp, and setting Course to null in ActiveSessionId in SessionManager before the backend starts saving the transcript and scheduling SessionSummaryWorker. The latter will get the saved transcript, invoke the LLM API externally, and save Session.SummaryText in PostgreSQL, as well as send summary_update and SessionEnded events to PresentationHub until the user requests replay using GET /api/sessions/{sessionId}/replay.

This decoupling enables the presenter to exit early, even if the LLM API invocation takes longer than needed. The replay path sends a single request in order to obtain the transcript, summary, and download links. Thus, this design keeps FR-REPLAY decoupled from the live session path, ensuring that post-session processing does not affect the presentation after the session ends. Even if the API call fails or times out, replay of the session is possible through slides, transcript, and downloading the artifacts, and generating summaries can be done asynchronously.

3.4 Functional and Domain Modeling

3.4.1 Use Case View

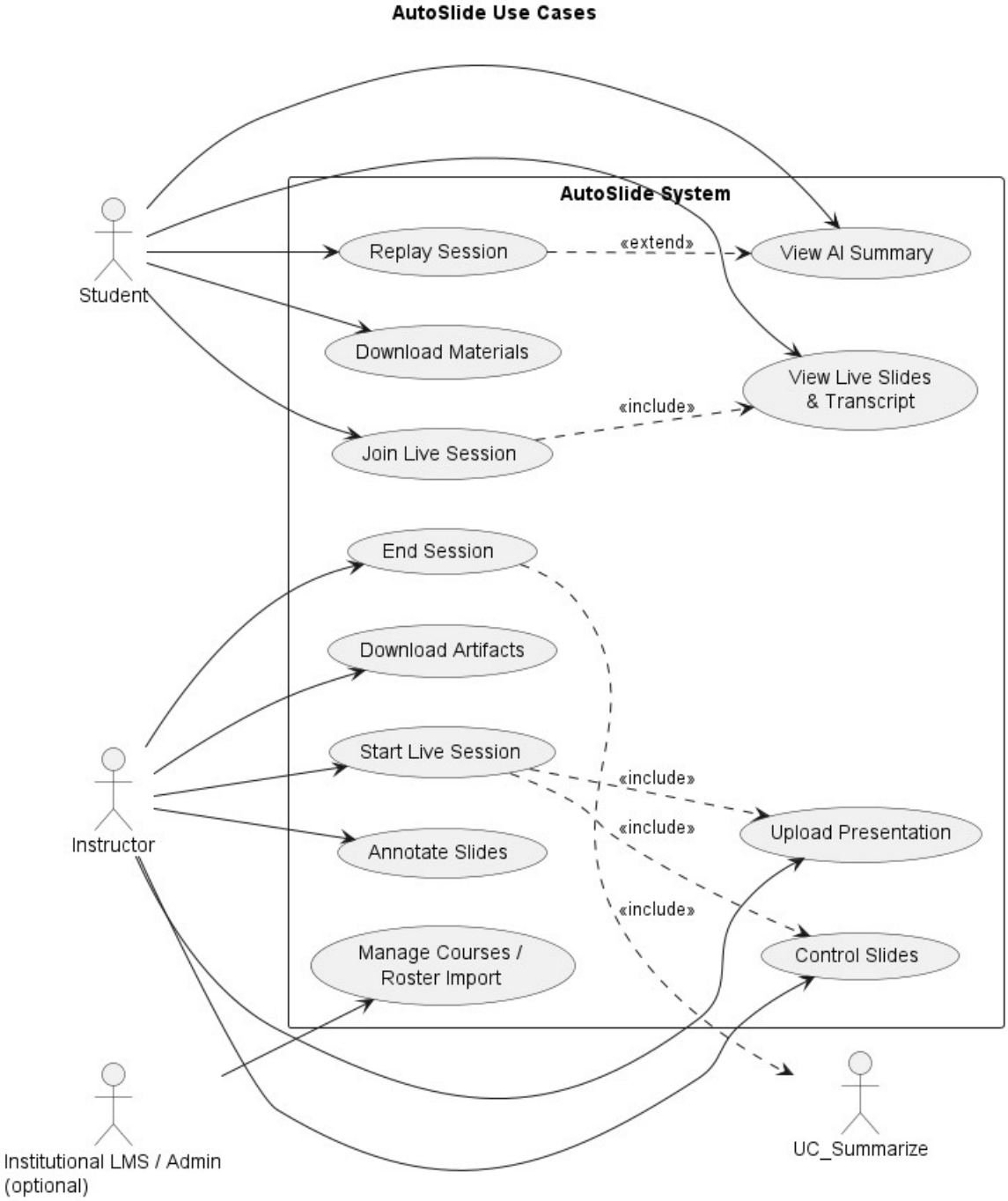

Figure 6 Use Case Diagram

The actor distinction for Instructor, Student, and an optional Institutional LMS or admin actor is important, as it breaks down the actions into Instructor actions, student review, and provisioning. This allows us to identify specific actors with different use case paths, rather than bundling the three actors into a generic portal use case flow.

The inclusion and extension relationships are significant in that Start Live Session includes both Upload Presentation and Control Slides, whereas End Session includes transcript summary generation. The Extend relationship for Replay shows that Replay extends the summary view. The result is a clear lineage between the use cases and FR-COURSE, FR-SESSION, FR-REPLAY, FR-AUTH, and FR-AI.

3.4.2 Domain Model

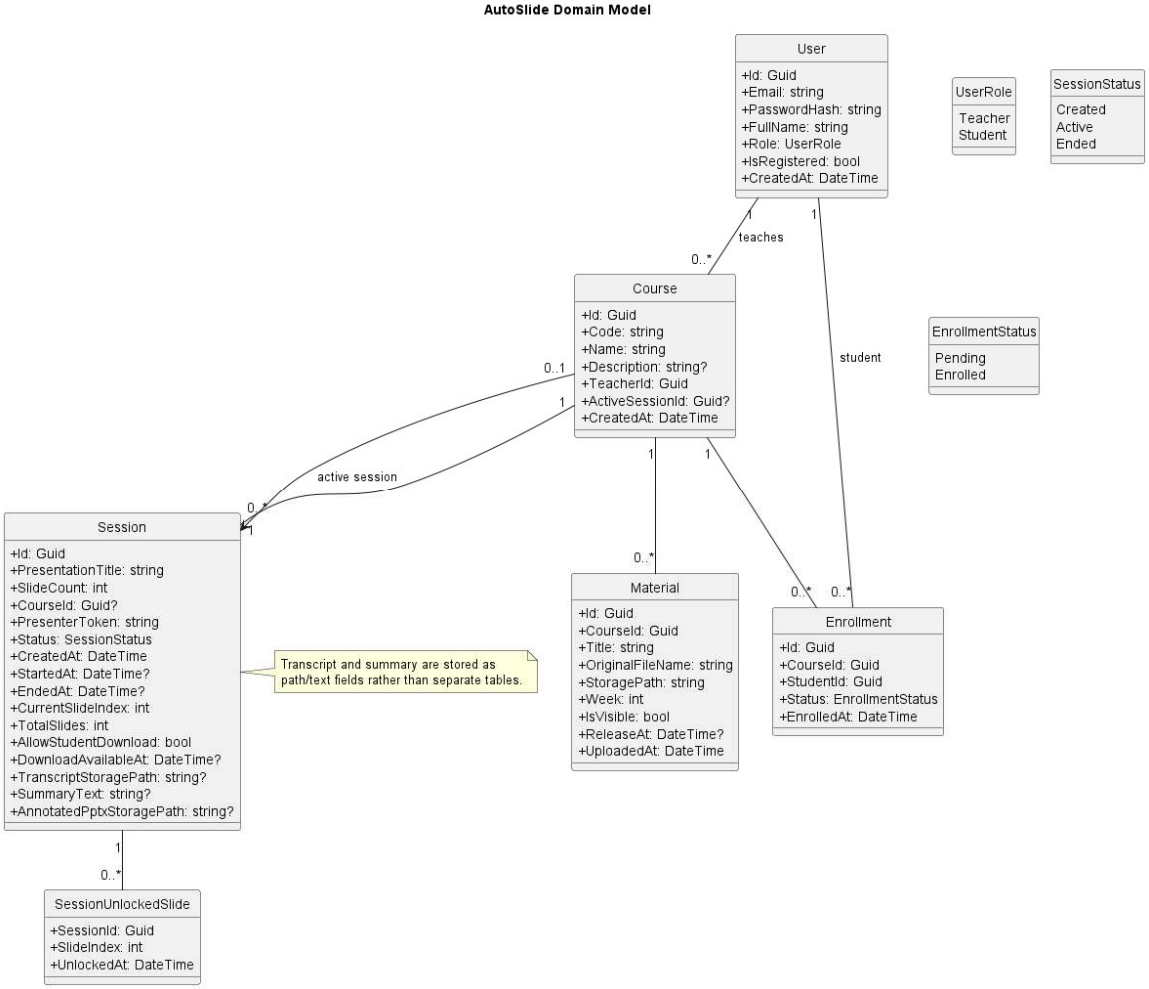

Figure 7 Domain Model Diagram

The domain model revolves around the User, Course, Enrollment, Session, Material, and SessionUnlockedSlide entities, as well as the UserRole, SessionStatus, and EnrollmentStatus enum types. The associations illustrate that the Course entity owns Materials and Sessions; the Enrollment entity connects a Student to a Course; and the Session entity has attributes including PresenterToken, CurrentSlideIndex, AllowStudentDownload, TranscriptStoragePath, SummaryText, and AnnotatedPptxStoragePath.

Clearly, this database design is mapped directly to FR-SESSION, FR-UNLOCK, FR-INK, FR-AI, and FR-MATS, because the Session table contains the live state, the SessionUnlockedSlide table tracks the progressive unlocking, and the Transcript and Summary columns contain post-session artifacts. Large objects, such as PDF presentations, annotated PPTX presentations, and transcripts, are stored in an object storage service, while metadata, ownership, and pointers are stored in PostgreSQL storage layer to authoritative relational store and provide runtime access controls. Storing transcript and summary artifacts as path and text makes the EF Core database design simple yet provides enough metadata for replaying and auditing purposes.

3.4.3 Session State Machine

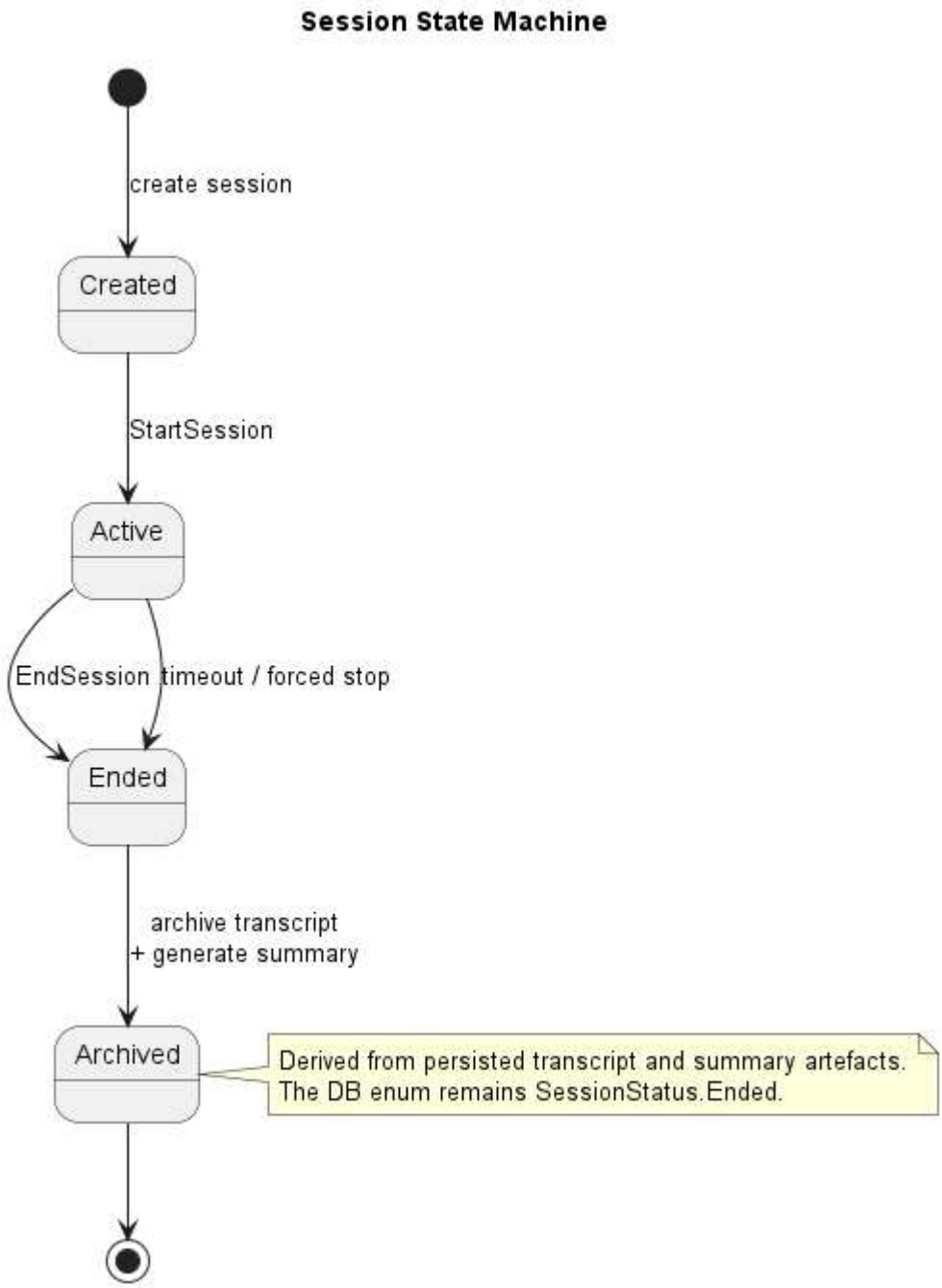

Figure 8 Session State Machine Diagram

For the state machine, the states are Created, Active, Ended, and then finally Archived once all of the artifacts, including transcripts, and summaries, are created. There are two possible transitions: StartSession and EndSession. If an end occurs due to a timeout or a forced end, the active session ends, enabling the backend application to close the live session. The final state of Archived is represented using a derived state rather than a value for SessionStatus to ensure persistence consistency. This works well in our architecture with respect to FR-REPLAY and FR-AI since the user interface could consider summarization a secondary task.

3.5 Design Integration and Architectural Trade-offs

The first rationale for using VSTO over Office's JavaScript API is that VSTO enables synchronous responses to slide changes and ink drawings via PowerPoint's COM library, whereas the latter does not reliably do so during live slide

shows. Second, the live viewer uses SignalR to receive push notifications whenever there are changes to slides or drawings, avoiding HTTP GET requests that may introduce delays and jitter.

In addition to the earlier design decisions, several operational factors also influenced the final architecture. First, managed services were used to store data rather than building a dedicated database in our infrastructure, which simplified the deployment process despite the strict FYP schedule, but at the cost of losing backup, availability, and scalability. Second, the SessionSummaryWorker is separated from the session stream and operates asynchronously, meaning that it can take any time to process and will not introduce additional latency between the API and presentation clients.

The most critical choice is who will keep track of the current state of slides and drawings and ensure consistency between presenters' screens. Rather than implementing state maintenance on the client side and syncing it with the server regularly, which can lead to desynchronization issues, our solution keeps the state at the backend and broadcasts updates to all viewers via SignalR. Although it makes us implement additional logic to maintain frame consistency within a slide, it ensures proper synchronization for newly joined and reconnecting users and replayed presentations.

From all of the aforementioned trade-offs, it is clear that our system prioritises real-time responsiveness, synchronization, and separation of responsibilities over the ease of implementation strategy.

4. System Implementation and Realization

4.1 Implementation Status Overview

Section 3 defined the architecture and boundaries of the AutoSlide components, but this section introduces the actual design of the prototype. The implemented version includes a PowerPoint VSTO add-in, an ASP.NET Core backend, React front-end web portal, SignalR real-time communication, Windows RTT voice transcription, and EF Core persistence using managed PostgreSQL databases and object storage. From a use-case perspective, this prototype allows for session initialization through PowerPoint for the presenter side, session state persistence on the back end, as well as browser access to live sessions and stored materials for students.

There were several key challenges in implementing this functionality, such as the application needed to support COM PowerPoint events, server-side session state, client-side browser rendering, and asynchronous post-session processing without hindering the presenter's workflow. For these reasons, this section illustrates the prototype as a workflow involving PowerPoint, ASP.NET Core, managed storage, and the web portal, rather than simply providing screenshots of individual components. The resulting prototype is fairly integrated and can be used to evaluate the solution.

4.2 System Interface and User Workflows

The interface follows an integrated LMS design, in which the screens for teachers and students are separated by role rather than by the technology used. By separating the roles, it ensures that each screen performs its respective functions, such as creating classes, attending online classes, reviewing recorded lectures, and downloading material.

4.2.1 Instructor Control Plane

The teacher workflow starts in the teacher dashboard, which serves as a pre-lecture control panel for creating courses, managing rosters, and scheduling materials. See Figure 9 below for the screenshot of the teacher dashboard and its use in pre-session administration.

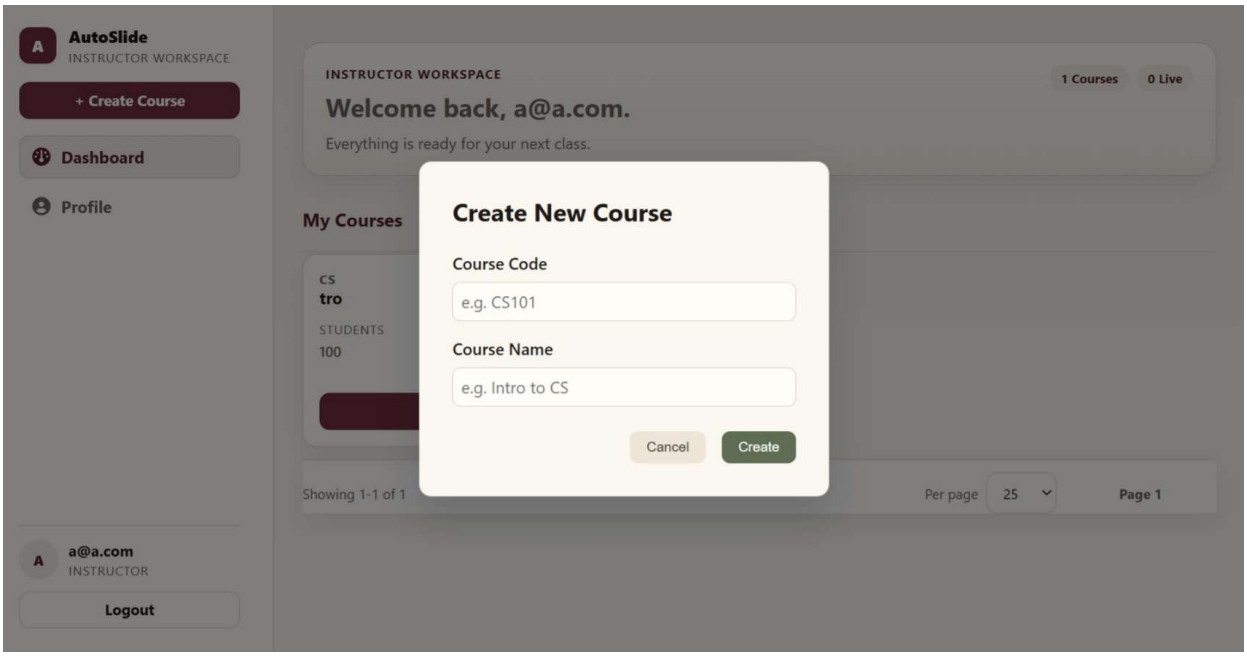

Figure 9 Create New Course Diagram

First of all, the lecturer creates a course to tie later enrollments, sessions, and releases of materials to it, thereby establishing the context for the entire process. Afterwards, the roster page helps manage the rosters of the courses

and import the students' access data before the start of the live session, thus creating an authorization boundary that will allow some students to have access to the course and its session. The same control panel enables the timed releases of learning materials. On the instructor's side, the materials view groups materials by course context and release time, allowing the instructor to upload supplementary files before or after the lesson in the same environment where they prepare lessons. Figure 8 below demonstrates how course materials are organized on the teacher's side for this process.

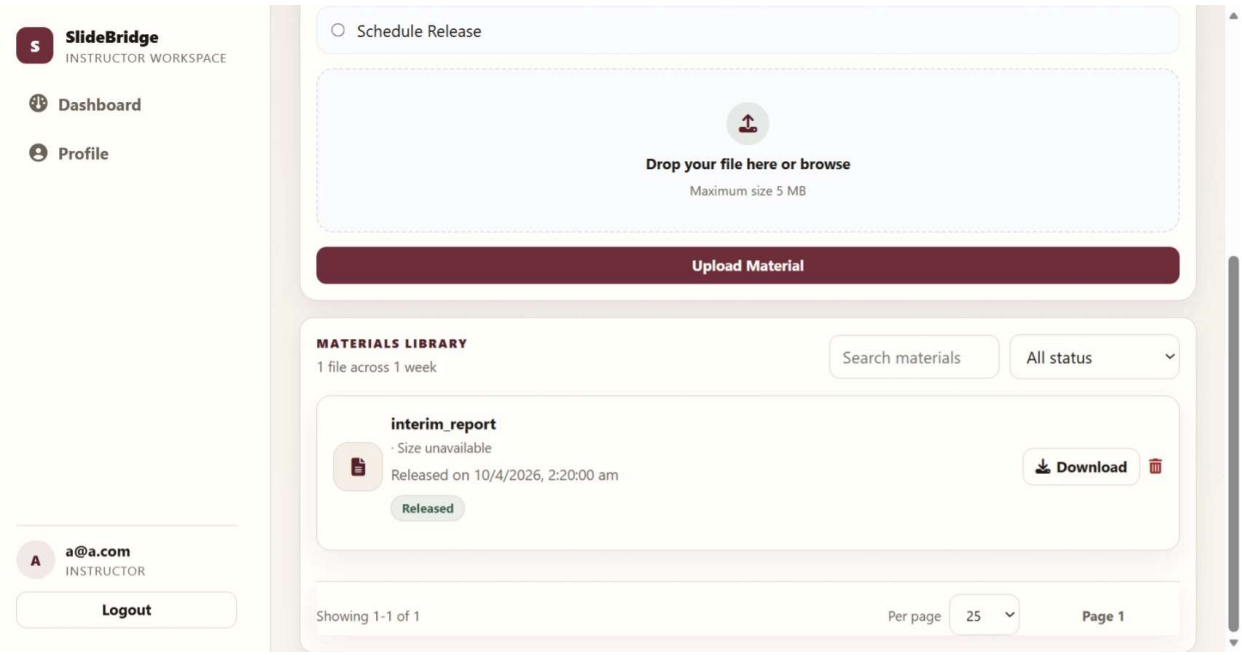

Figure 10 Instructor Manage Material Page

Upon completion of this stage, the control plane moves to PowerPoint via the VSTO ribbon, where the presenter begins sharing the screen from within PowerPoint. Refer to Figure 11 below to view the ribbon and locate the broadcasting controls.

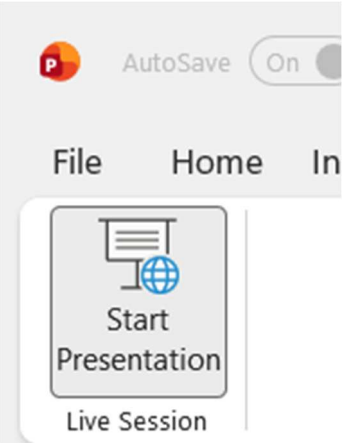

Figure 11 Add In Ribbon - Start Presentation Button

4.2.2 Student Web Portal

The learning experience starts on the dashboard, where only the courses the student is enrolled in are displayed, with a status indicator showing whether a live session is currently streaming. As shown in Figure 12 below, this is a

dashboard view from the learner's perspective that highlights the existence of a course membership and a live session in progress.

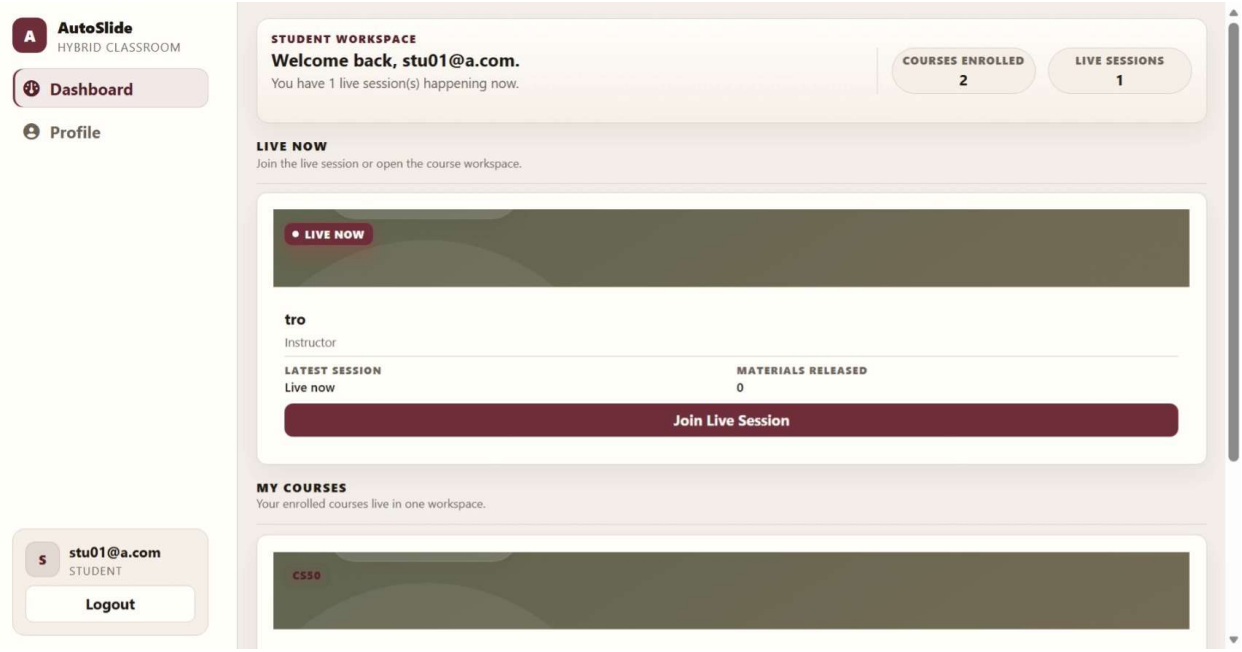

Figure 12 Student Dashboard with Live Session

During an active live session, the learner enters the live viewer from the dashboard. Inside the live viewer, the learner navigates the synchronized slide stream, views the transcript data in the same viewer and can only navigate the part of the slide stream that has been unlocked for view as instructed by the teacher at that specific time of the lecture. The live viewer, from the student's perspective, is shown in Figure 13 below.

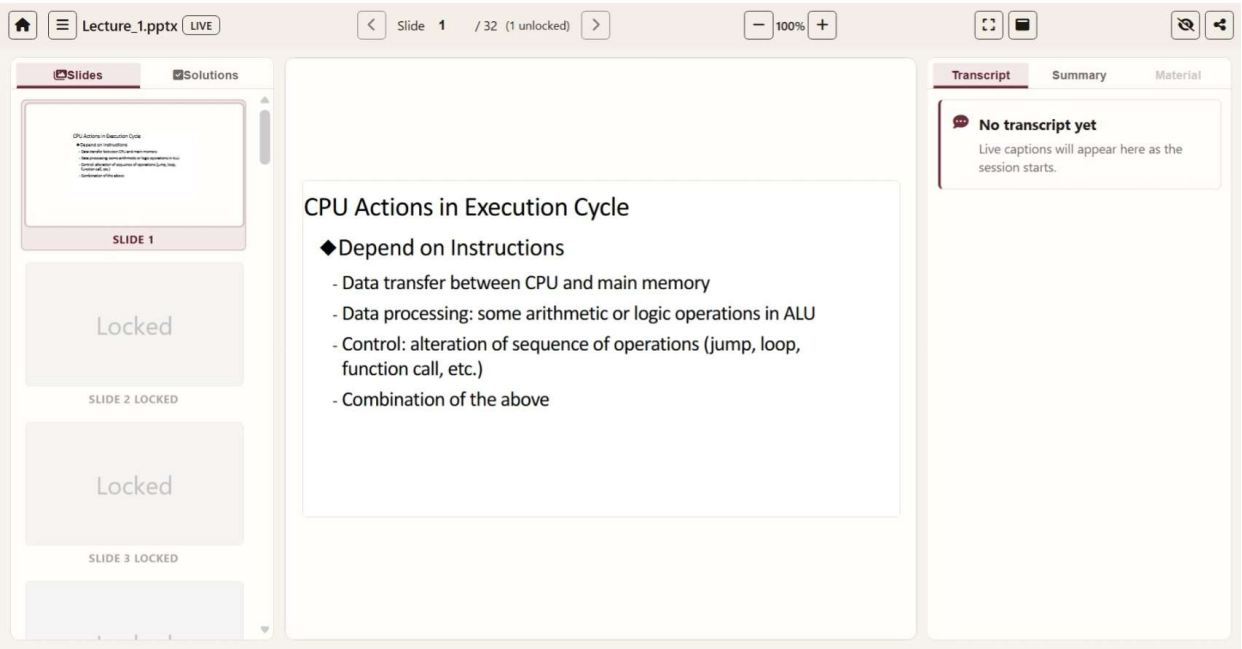

Figure 13 Student Live Viewer Interface with undisclosed slide locked

4.2.3 Session History and Replay Interface

Once the session is complete, the learner can enter the course and find its session history within the dashboard panel. Such history view gives an opportunity to replay in an organized way instead of searching separately for transcripts or export files, as well as any other support material.

In the replay window, the learner can view the recorded session context and download available artefacts from the session recording. Figure 14 demonstrates the session history and replay interface that allows combining slides, transcripts, AI-generated summaries, and downloadable artefacts into one session record.

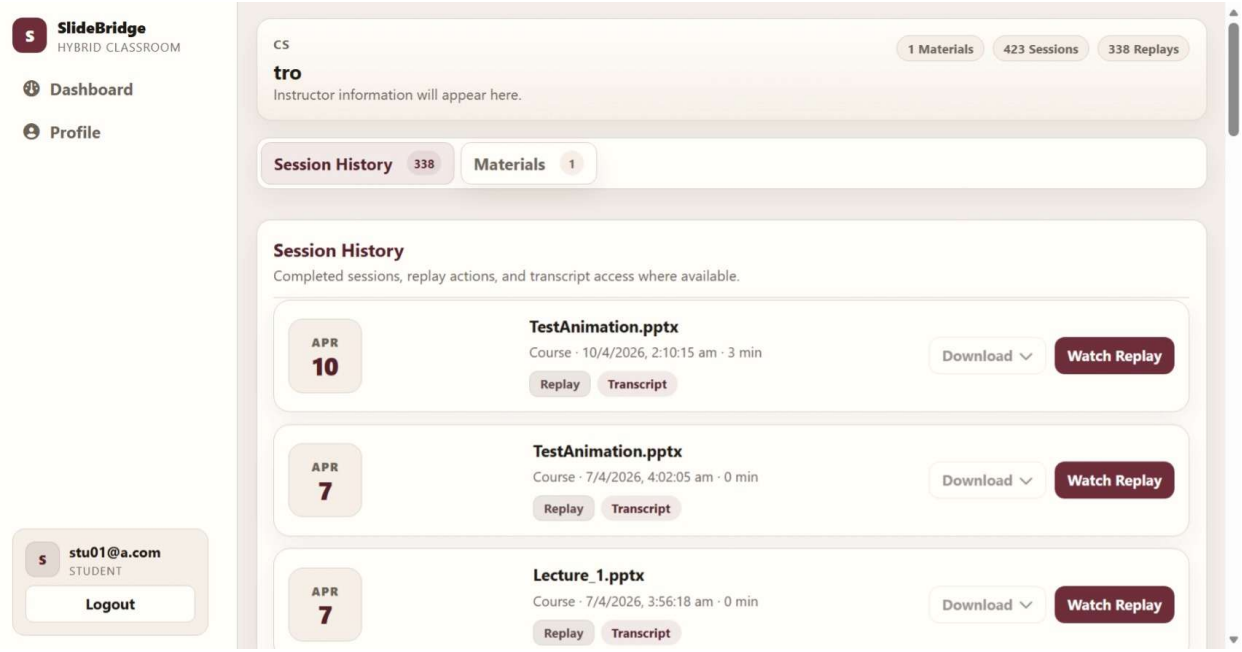

Figure 14 Student Course Session History for replay and download material

Thus, the workflows of Section 4.2 mainly address Deficiency 1 by cutting down on the overheads related to manual publication of learning material, solve Deficiency 2 using controlled release of visible slides, and partially address Deficiency 3 providing integration of live viewing, transcript access, materials, and replay into one role-aware portal.

4.3 Core Functional Modules Realized

Whereas Section 4.2 discussed user actions and experiences, this section illustrates how the implemented modules facilitate these processes through the desktop, server, and browser tiers. The key anchors for implementations connecting the PowerPoint Add-In, backend services, and Viewer components are provided in Table 2.

Module	Key endpoint or hook	Purpose
Session delivery	POST /api/sessions/start	Registers the session and initiates backend processing for live delivery.
Session delivery	SlideShowNextSlide COM event	Captures PowerPoint slide advancement from the VSTO side.

Live synchronization	PresentationHub.SlideUnlocked SignalR event	Broadcasts authoritative slide and unlock state to connected viewers.
Replay delivery	GET /api/sessions/{id}/replay	Serves consolidated replay data for post-session review.
Ink persistence	POST /api/ink/{slideIndex}	Persists captured ink data associated with session playback state.
Transcript pipeline	POST /api/transcript	Stores live speech fragments for replay and summarization.
Summary generation	SummaryWorker.RunAsync	Triggers asynchronous LLM-based post-session summarization.

Table 6 Selected implementation anchors across the AutoSlide stack

4.3.1 Session Creation and Real-Time Slide Synchronization

Fig 11 show creating the live session involves clicking the VSTO add-in ribbon to share the presentation from PowerPoint. In terms of runtime, the presentation is uploaded, the live session is registered, and the presenter is added to the SignalR session group, while the student receives session information and connects to the live stream. As described above, there will be no delay in starting the lectures, as file uploads and conversions occur automatically without waiting on the side until the session becomes accessible to students. Fig 6 after creation of session an QR code will be available to students for join sessions.

Figure 15 QR code and join session message box

Navigation events are fired by the live session controller, which publishes them via the SignalR hub rather than making poll requests in the browser. This approach allows overcoming Deficiency 1 as live streaming takes place as part of a synchronization process without a post-session publication, and Deficiency 3 as session-based presentation and viewing control are implemented.

4.3.2 Progressive Unlocking and Access Control

The solution provided is server-based, where the client-side only receives updates from the server on what to display to the audience during the progressive unlock process. Fig 13 shown every time the presenter unlocks a new slide, the server captures the current slide position and the unlock slide boundary, then pushes them to the connected viewers via SignalR.

This means that, unlike in the original version of the system, where the students decide on their own what slides need to be revealed, in this case, the client will receive information about unlocking status from the backend, ensuring that even latecomers have access to the correct sequence of slides, just like in the original session.

4.3.3 Ink Capture, Persistence, and Export

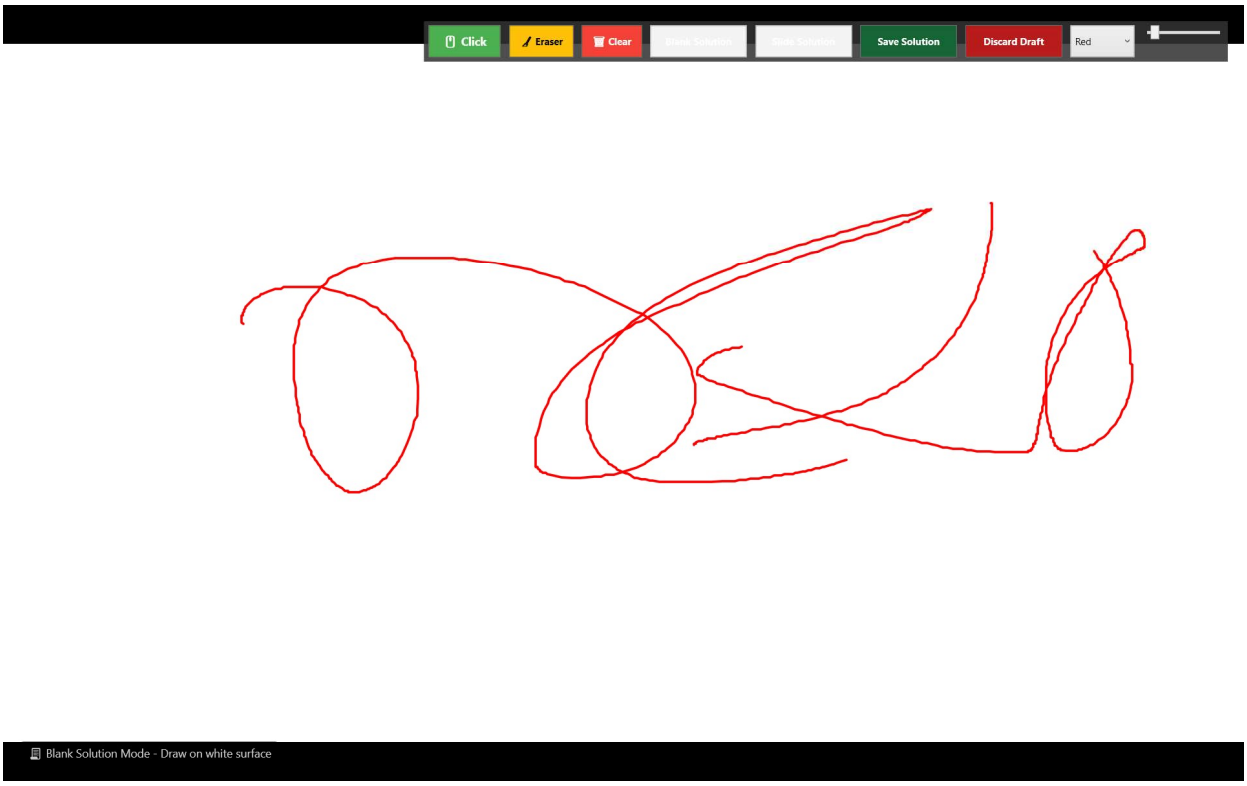

Figure 16 Ink capture overlay where instructor can draw solutions

Fig 17 show a critical implementation artifact capture because the live teaching annotations are likely to include explanations not included in the initial deck of slides. The system design deliberately treats ink capture and playback as first-class artifacts of the session instead of simple visual effects.

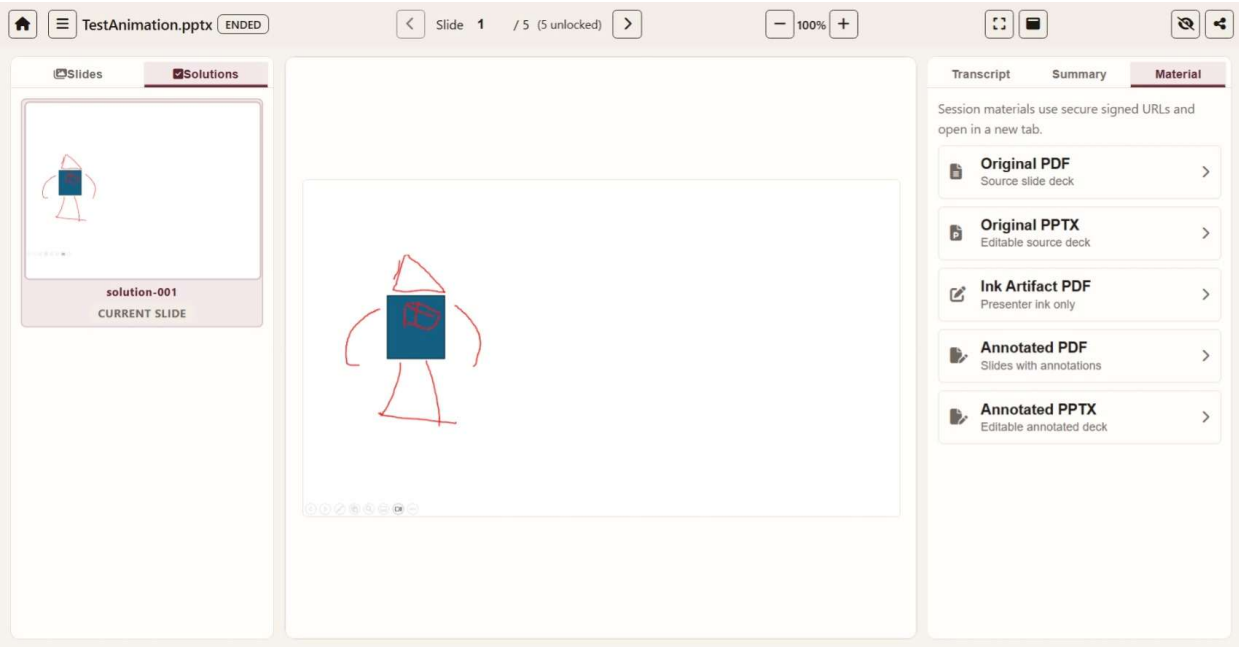

Figure 17 Saved solution for student review

When the session is active, Fig 18 show saved ink is sent over the overlay and live delivery channels and persists so it can be reconstructed during replay and export operations. In addition, the storage service retains annotations and corresponding artefact paths required for downloading during replay and exporting with annotations. When the session end student can use the annotated solution slides to aid their study.

One of the difficulties encountered in this scenario is slide tracking and synchronization, as there is a requirement to track more than just the slide number due to animated slides. In such a case, multiple visual frames may appear within a single PowerPoint slide, making one-to-one mapping of presentation state and replay state impossible. As part of the compromise design approach, the system design gives preference to the stability of live synchronization and annotation persistence instead of exact replication of every visual frame at each point of the animation. This means that baked ink collected through the overlay channel gets reused for all frames in the animated slide, while in a separate slide solution, it would get synchronized one-to-one with all intermediate animation frames.

4.3.4 Live Transcription and Post-Session Summarization

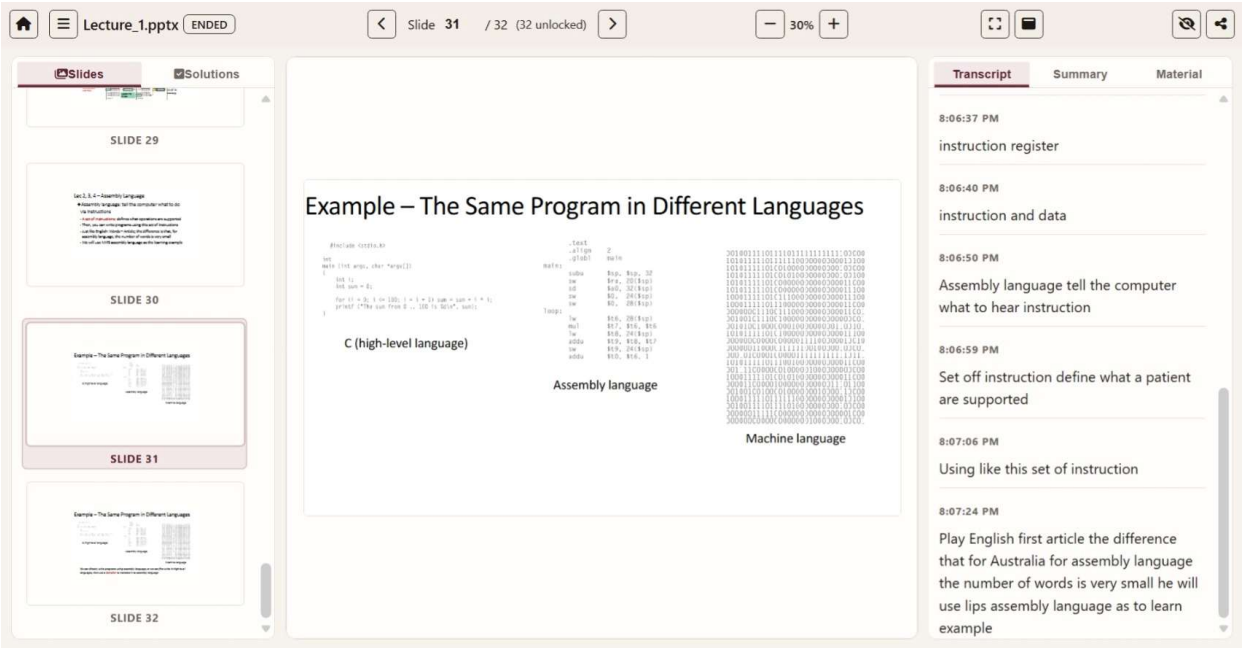

Figure 18 Side panel with Live Transcription

The live transcription feature uses the WinRT speech library in the instructor module and is distributed through the transcript window in the web viewer module. The transcript chunks will be created at runtime during the live lecture session as shown in Fig 19, transferred to the backend server for storage, and used later for replay and summary generation.

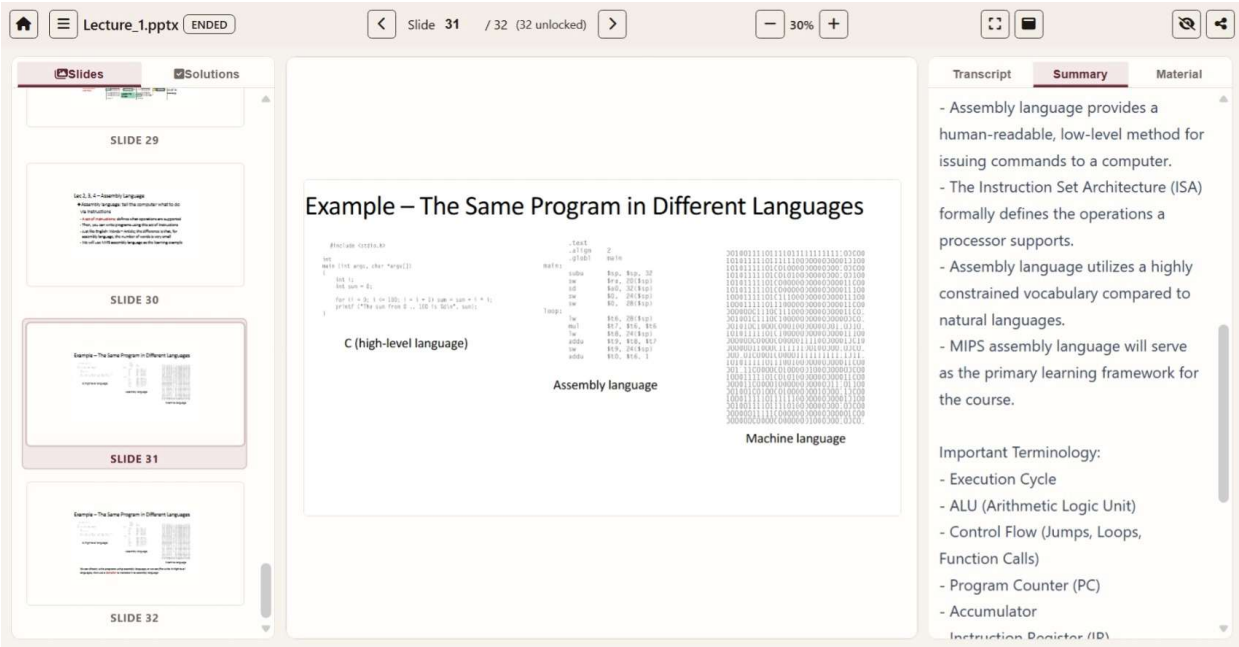

Figure 19 AI supported Transcript Summary

The summarization process, in turn, is not included in the pipeline of the live lecture session by design. After completing the live lecture session, the backend module saves the transcript and makes it accessible via the playback interface. Summaries are generated only when a user requests them and are handled by the LLM provider. It is important to note that such an approach helps save resources, as summaries are generated only when the user requires them as shown in Fig 20. Thus, it is possible to say that the current architecture solves Deficiency 3.

4.3.5 Materials Release and Artefact Delivery

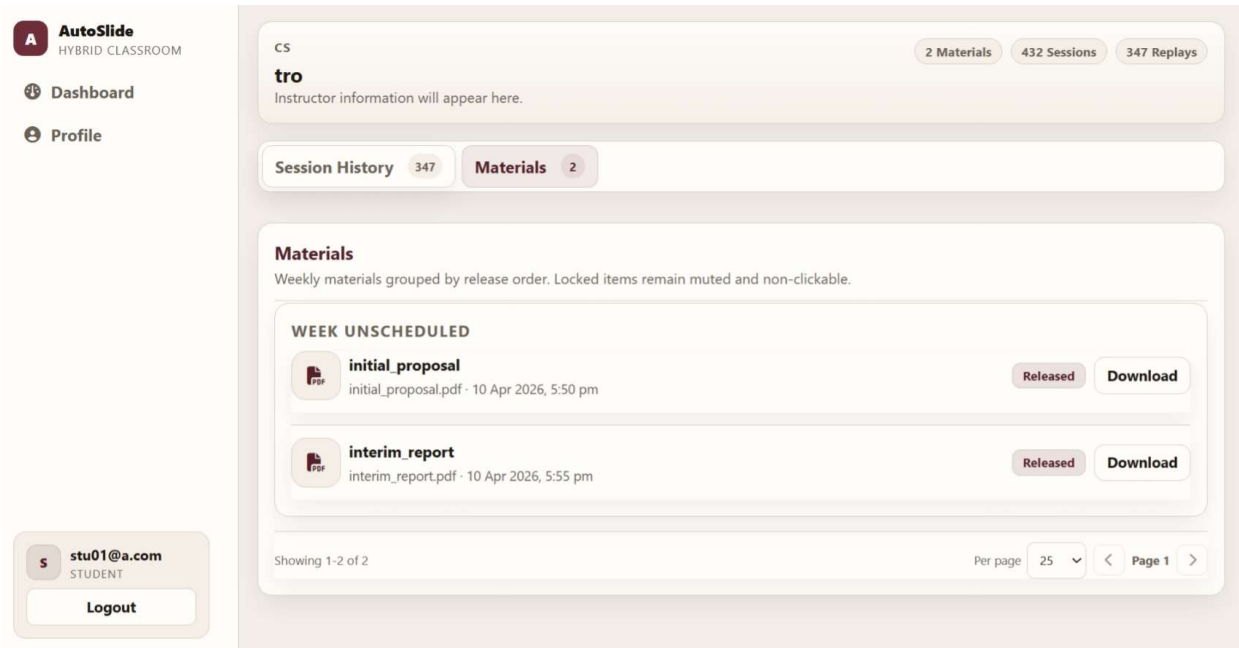

Figure 20 Student Course Materials page

The materials release feature is facilitated through the course-related record and authorization model employed in the sessions and enrollment process. From the instructor's point of view, materials are scheduled for release.

Artifacts delivery takes the same approach to deliver session-specific results like transcripts, annotated files, recordings, and others. By utilizing these artifacts via back-end and portal, the realization not only eases the follow-up distribution procedure but also retains consistency with respect to content taught, content captured and what students may eventually refer to. Hence, this improves Deficiency 1 and Deficiency 3.

To summarize, it can be concluded that the realizations described in Section 4.3 meet all three deficiencies, but to varying degrees. Synchronization and ink persistence mainly address Deficiency 1, the progressive unlocks feature resolves Deficiency 2, and the realization of transcription, summarization, replay, and controlled artefacts delivers Deficiency 3.

4.4 Technical Challenges and Design Trade-offs

Some non-trivial design choices were made during development, which will be discussed here to explain why the particular combination of technologies was chosen over a simpler alternative.

First, VSTO rather than Office JavaScript was used to leverage PowerPoint's event granularity and ink capabilities for synchronous annotation capture. This decision improves responsiveness but locks the instructor to the Windows platform and VSTO deployment scenario.

Second, SignalR, rather than HTTP polling, was used to provide a smooth live session experience even when rapidly navigating between slides. However, SignalR introduces additional complexity in handling reconnections, updating viewer states, and maintaining proper group membership for both desktop and web clients.

Third, slide tracking and synchronization imply additional complexity in state management compared to simply aligning by slide number. Namely, because there may be many intermediate visual states within a single PowerPoint slide, particularly when animations are present, the solution cannot always assume there is one slide state per replay slide without losing accuracy. For this reason, a compromise was made in favor of more stable synchronization and retained annotations' visibility instead of perfect frame-based replay. As a result, baked ink captured in the overlay window is associated not only with one particular slide frame but also with its neighboring frames, unlike a separate slide solution, where this relation is one-to-one.

Fourth, PostgreSQL and object storage are employed in the prototype to store all generated artifacts, thus reducing deployment complexity during development time but preventing large-scale optimizations in favor of a validated prototype.

Fifth, transcript summarization is done in a separate task to preserve performance, but at the expense of introducing some AI-related dependencies. Specifically, the presenter's experience is protected from AI latency, but the quality of summary will depend significantly on transcript quality, LLM capabilities, and future refinements.

In summary, it becomes clear that the presented solution prioritized presenter experience, artifact integration, and session orchestration over production hardening and other aspects. Therefore, the system should be viewed as a substantial prototype for testing rather than a full-fledged institutional solution.

5. User Study & System Validation

The section includes the results of the controlled experiment that was carried out via an anonymous survey involving 50 participants – 40 students and 10 lecturers – who worked on three cases using the AutoSlide software. The evaluation is made considering the information obtained during their performance, surveys, and observation. Hence, the assessment takes place through empirical analysis and not only on the design level.

The experiment has been performed according to the validation strategy outlined by the project's supervisor. In particular, the plan involved about 50 subjects, including students and teachers. In the final dataset, users are represented across three key workflows in AutoSlide: Basic Live Lecture, slide Unlock with Ink annotation, and Session Replay and transcript Summary.

Measure	Value
Total participants	50
Students	40
Instructors	10
Scenario A - Basic Live Lecture	15
Scenario B - Unlock + Ink	20
Scenario C - Replay + Summary	15

Table 7 Test Subject and test break down

5.1 Study Design and Participant Demographics

The validation process was performed using a usability methodology based on walkthroughs, in which each participant performed one lecture session scenario and answered a follow-up questionnaire post-session. The instrument consisted of four evidence layers:

- Part A consisted of the results of task accomplishment,
- Part B collected scaled evaluations of usability and usefulness,
- Part C obtained qualitative reflections and the observer section recorded hesitation, assistance, and disruptions.

The Part B questions were answered on a five-point scale, with higher numbers indicating stronger agreement or greater usefulness. Such a setup enabled triangulation for evaluating performance, experience, and external observations within a single validation session.

The scenarios were manually distributed among participants to avoid a random distribution that would make it difficult to achieve a balanced number of participants across the three workflows. Role information was kept.

The profile of participants was biased towards students, since the viewer workflow was intended to be the primary use of the system during lectures. Still, there is some methodological significance in having ten instructors participating in the test since there are AutoSlide features useful specifically to instructors, namely, publication control and artifact generation.

Participants' anonymity was preserved by using unique participant IDs rather than any other form of identification. Such an approach ensures that all reports will be uniform and that no confidential data will leak, while still allowing comparisons between groups of participants and scenarios.

5.2 Task Design and Task-Completion Results

The task design followed the lecture lifecycle structure as implemented in AutoSlide. Firstly, join-and-setup tasks were tested to see whether the subject could access either the viewing or presenting flows, initiate the lecture session correctly, and determine the session's status.

Secondly, the live-session tasks evaluated the efficiency of slide transitions, slide unlock clarity, slide ink location, proper animation functions, transcript visibility, and interface clarity.

Thirdly, the replay-and-artefacts tasks analyzed the subject's ability to initiate the recorded lecture session, evaluate the content's fidelity during live session versus replays, view the transcript and summary output, identify the download zone, and retrieve artifacts.

Finally, this research captured general completion results to measure participants' ability to perform their assigned scenarios independently and to what extent that process would be successful in a real-life educational scenario.

Overall, aggregated data from Part A showed that the task-completion category scored the most success, with an overall success rate of 89.0 percent in all applicable responses. The success rate for join-and-setup tasks was 86.6 percent, and that for live-session behavior was 81.5 percent. Finally, the success rate for replay-and-artefacts was a low 76.2 percent, which indicated that the retrieval process post-session was not as successful as the live delivery process.

Part A subsection	Applicable responses	Successful	With issue	Success rate
Join and setup	194	168	26	86.6%
Live session behavior	336	274	62	81.5%
Replay and artifacts	147	112	35	76.2%
Overall task completion	100	89	11	89.0%

Table 8 Task completion with highest satisfaction

Based on this pattern, the implementation of the system's architecture already supports reliable end-to-end operation of the main lecture pathway but replay and artifact extraction are still problematic areas requiring improvement. This is consistent with the implementation history covered elsewhere in this report, especially in transcript generation, AI summaries, and certain slide animation cases.

5.3 Quantitative Analysis: Usability and Perceived Value

Based on the results of Part B, it can be noted that high usability is not restricted to the interface but is also found in the usefulness of the whole process in lecturing. The highest means were recorded for Pain alleviation, productivity, and Comparative value with 3.85 points out of 5. Third highest was obtained by Usability with 3.75 points. Based on these findings, it can be said that the test subjects considered AutoSlide useful despite its shortcomings in design.

A Likert scale of 1 (lowest), 2, 3, 4, and 5 (highest) was used to rate agreement with the statements in the questions, as well as the usefulness of particular features and functions.

On an individual item basis, some of the best performing items include B26: "AutoSlide solved the problem of slides being unavailable during class lectures." It got a mean score of 4.02 out of 5. Other items with good performance were B16: "Changes in slides happened swiftly" and B31: "AutoSlide would increase my productivity", scoring 3.94.

The same good performance was shown by B28: “AutoSlide helps in reviewing class after lectures” and B21: “Replay matches live classes”, scoring 3.92. These results support the research thesis that AutoSlide makes lectures accessible, maintains instructional state, and eliminates fragmentation in live and post-class deliveries.

The worst results were recorded for AI summary features and transcripts. The lowest mean came from B25: “The AI summary was useful,” with a score of 3.26. Other low-meaning items included B19: “Live captions/transcripts were useful”, getting a mean of 3.32 out of 5 and B20: “Interface was easy to use when delivering lectures”, scoring 3.60.

Rating group	Mean	Median	Interpretation
Baseline pain points (Students)	3.71	4.0	The baseline pain points on the students' side are very much evident within the population in question
Baseline pain points (Instructors)	3.72	4.0	Instructor-side process-related pain is also very evident.
Usability	3.75	4.0	Usability in general is positive, but there is still room left for improvement.
Live features	3.67	4.0	Live delivery is excellent, but subtitles and arrangement could do better.
Replay and artifacts	3.71	4.0	Replay was appreciated, but the usefulness of summaries made the score for the block lower.
Pain alleviation and productivity	3.85	4.0	This category offered the best evidence for the practical utility of the system.
Comparative value	3.85	4.0	Participants regarded AutoSlide as more valuable than static post-class slide sharing.

Table 9 User experience and interpretation

In conclusion, we can derive a crucial statement regarding the obtained findings. It is clear that the AutoSlide technology demonstrates its dominance in addressing issues related to availability, persistence, and search capabilities. Nevertheless, it is worth mentioning that when it comes to handling intricate processes implying significant semiotic competencies or enhanced balancing of the user interface, such as live transcription, intelligent summarization, and multiple panels with many objects in each one of them, the system turns out to be relatively inefficient.

The data gathered from the observer further confirms this analysis. Overall, across 50 test runs, 13 individuals appeared uncertain, 7 needed assistances, and 4 experienced a critical failure. While none of these numbers contradict the general success of executing the required procedures, they indicate that the current version of the technology has enough functional inconsistencies to require a separate post-validation optimization phase rather than immediate implementation.

5.4 Qualitative Feedback and Observer Notes

The qualitative responses explain why some quantitative measures were lower and provide concrete evidence for the future work agenda proposed in Section 6. In general, there were seven main themes among the qualitative responses: transcription accuracy, collaboration on the shared whiteboard, quality of generated summaries, accuracy of animations, vision-based AI explanation support, attendance tracking, and multimodal question-answer interactions. Multimodal question answering was particularly important because multiple participants expressed a desire for interaction between slide content and handwritten notes.

There is clearly a divide between opinions regarding AutoSlide's current capabilities versus areas for improvement. Participants found that it offered the possibility of synchronization between slides, preservation of instructor-ink, recording for later replay and collection of all the content in one workflow. However, several participants also

emphasized that the system was still too focused on content delivery rather than on real-time classroom interaction. As a result, the topics of collaborative whiteboarding and multimodal interaction are frequent in both the quantitative and qualitative data and are central to future system development in Sections 6.2 and 6.4.

There is also an important difference between the instructor responses and those from students. The former tends to mention features related to administration and workflow, namely attendance tracking and operational feedback, while the latter tend to ask for more interaction and explanation. The difference between these role-related expectations is analytically interesting, as it implies that the future refinement effort should not be regarded as a generic improvement of the current system, but rather as two different directions: workflow analytics for instructors, and interactive learning tools for students, presented in Sections 6.1 to 6.4.

In conclusion, this study provides strong formative evidence for AutoSlide as an RT-HLS system. First of all, the evidence confirms that it can successfully address at least some of the baseline problems with slides, ink, and post-classroom review. On the other hand, this study also identified a specific refinement agenda focused on transcription, summaries, replaying, collaboration, attendance, and AI explanations. Thus, the findings serve as an empirical foundation for the future work agenda outlined in Section 6.

5.5 Threats to Validity

The weaknesses of the study can be understood through the definition of the two types of validity – internal and external validity, according to conventional literature on experimental and quasi-experimental evaluation [18], [19].

5.5.1 Threats to Internal Validity

Regarding internal validity, there are at least four factors worth noting. Firstly, random assignments were not performed in the present research. To balance the distribution of scenarios, they were manually allocated to three workflows. Such an approach ensured operational practicability but, at the same time, opened up the possibility of threats posed by pre-experimental characteristics of different individuals or user roles, which influence the outcome independently of any intervention [18], [19].

Secondly, a guided walkthrough approach was adopted to conduct the evaluation. Participants were doing their work under supervision and having complete information on what they needed to do; therefore, their results may have been inflated due to the absence of supervision [18], [19].

Thirdly, the evaluation incorporated both task records and observers' notes, as well as post-session questionnaires. While such a combination increases the credibility of the research, the data obtained from the questionnaires can be affected by instrumentation issues, i.e. interpretation and response biases, as well as novelty effect, particularly with regard to subjective assessments of usefulness and value [18], [19].

Finally, instability occurred in some of the evaluated features during further system development, such as transcript generation, AI summary, and handling certain animation events. As a result, the corresponding feature may have received a lower rating because of instability in the prototype, which did not affect its performance in the final product [18], [19].

5.5.2 Threats to External Validity

With regards to external validity, the first issue is limited generalizability. Despite being conducted with a relatively large number of participants, 50 altogether, the sample was recruited under controlled conditions related to the current project and consisted largely of students (40 students vs 10 instructors). Therefore, this sample is sufficient for formative validation of the viewer's workflow but does not provide much basis for making assumptions about the deployment of the system by instructors or its wider institutional implementation [18], [19].

Another issue is that participant demographics were mostly recorded on a basic role-based level. Specifically, detailed information regarding such factors as academic discipline, study year, teaching experience, as well as the experience of working with hybrid-learning tools, was not gathered, thereby restricting the scope for analyzing representativeness and subgroups [18], [19].

Furthermore, the evaluation was conducted in a controlled, guided walkthrough mode. Participant behavior in this mode might differ from that observed under natural conditions, as actual class sessions involve many factors, such as time pressure, potential interruptions, varying connection and presentation quality, and long-term use [18], [19].

Moreover, the interaction of selection and experimental treatment cannot be ruled out. Namely, all participants, scenarios, and the experimental environment were selected with respect to the requirements of the current project; therefore, the observed benefits might not necessarily be applicable to other universities, subjects, lecture formats, or student audiences [19].

Finally, although all three testing scenarios are consistent with typical AutoSlide use cases, some uncommon lecture formats and pedagogical approaches might yield interesting application patterns not covered in the above scenarios.

Overall, the present study can be regarded as a formative validation of the developed prototype under controlled conditions rather than as conclusive evidence of its operational readiness for actual use. However, the results provide strong grounds for stating that AutoSlide works effectively and shows great potential under the tested scenarios [18], [19].

6. Future Work and Post-Project Refinements

The roadmap for future work is directly informed by qualitative and observational insights from the user study, including the need for collaborative interactions, improvements to transcript and summary generation, replay accuracy, and attendance analytics. The roadmap takes into account the implementation details described in section 4, but it must be constrained by the results of the user study, which include poor transcript and summary quality, issues with UI discoverability, and a lack of collaboration.

By this stage in the tool's development, almost all the features we planned to implement in the second half of the process had already been implemented and evaluated by 50 participants. This means the main challenge at this stage is not to implement new functionality, but to ensure the system works reliably and that there is no ambiguity about what needs to be done in each case.

This involves six concrete areas of development: attendance analytics, interactive collaborative whiteboarding, artificial intelligence-powered explanations based on visual input, improved transcription and minimization of distractions, enhanced summarization as a separate learning resource, and improved animations for complex slide sequences.

6.1 Attendance Tracking Integration (Post-Validation Enhancement)

Instructor feedback suggests greater integration is needed, tying live delivery to administrative operations for attendance tracking. Post-validation improvements will allow AutoSlide to surface live session attendance analytics, including session start times, duration of presence, and replay access, in a structured format. The view will track authentication status and push the data into the course dashboard or the institution's LMS via an API, making it possible to see who joined the class, how long they were present, and whether they accessed the artifacts in question without needing additional attendance management.

6.2 Collaborative Whiteboard and Split Teacher-Student Canvas

Consistent qualitative responses indicate a desire to move from mere passive consumption to collaborative problem-solving through the classroom whiteboard, where students can attempt their answers in real time. Next step is to add collaboration to the whiteboard view by splitting it into two zones – one for instructor ink and one for students' contributions. It means extending the existing ink overlay functionality to include role-based permissions, conflict detection and resolution mechanisms, and the ability to record replay information for both instructor and students' attempts.

6.3 Vision-Based AI Explanation of Ink and Course Material

Student requests reveal a desire for more vision-based AI that will be able to interpret teacher ink and explain it in context along with course material. It suggests adding another module to the application to extend its existing

capabilities. Specifically, a combination of multimodal models should be added to the application's architecture to generate step-by-step explanations of handwritten diagrams based on the slide's content. It is directly aligned with the existing summarization pipeline and solves the AI-assisted learning and course materials interaction issues.

6.4 Multimodal Q&A Integration (Post-FYP Future Work)

The user study highlights that while AutoSlide solves the problems of access and artifacts, it does not offer rich two-way communication. Specifically, students and teachers both ask for whiteboards, splitting layouts with whiteboards and also AI capability in interpreting handwritten artifacts together with slides. This user study input maintains multimodal Q&A as valuable post-FYP work.

For this purpose, the viewer application will evolve from simply displaying the content to an interactive medium where users can pose multimodal questions. The drawing-based interface will allow users to ask multimodal questions by drawing directly on a whiteboard slide and attaching their questions to particular parts of the teacher's lecture slides. Using a multimodal Q&A engine that integrates visual AI capabilities and image recognition technologies will interpret the diagrams on the slide and the handwriting on the whiteboard and provide reasoning behind them.

As mentioned above, developing such capabilities relies heavily on multimodal modelling and vision-based AI and, therefore, is deferred to post-FYP work.

7. Conclusion

AutoSlide moved beyond the concept of proof of concept to become a nearly production prototype. The evaluation consisted of walkthroughs and a controlled study of 50 users. This research produced the necessary infrastructure and an LMS-style material publishing flow, secure slide-level permissions, ink persistence/export, replays, and a full pipeline for summarizing presentations after sessions. All of these accomplishments directly addressed the shortcomings described in the introduction: the effort required to manually publish slides, the lack of presentation-controlled progressive disclosure, and the disjointed AI interactivity features. This project turned a conventional slide presentation into an AI-enhanced teaching workflow while adhering to privacy and equity requirements.

References

[1] A. Raes, L. Detienne, I. Windey, and F. Depaepe, “A systematic literature review on synchronous hybrid learning: Gaps identified,” *Learning Environments Research*, vol. 23, no. 3, pp. 269–290, Oct. 2020, doi: 10.1007/s10984-019-09303-z.

[2] I. C. Noppe, “PowerPoint presentation handouts and college student learning outcomes,” *International Journal for the Scholarship of Teaching and Learning*, vol. 1, no. 1, Art. no. 9, Jan. 2007, doi: 10.20429/ijsotl.2007.010109.

[3] P. Chandler and J. Sweller, “The split-attention effect as a factor in the design of instruction,” *British Journal of Educational Psychology*, vol. 62, no. 2, pp. 233–246, Jun. 1992, doi: 10.1111/j.2044-8279.1992.tb01017.x.

[4] E. Ball, H. Franks, J. Jenkins, M. McGrath, and J. Leigh, “Annotation is a valuable tool to enhance learning and assessment in student essays,” *Nurse Education Today*, vol. 29, no. 3, pp. 284–291, Apr. 2009, doi: 10.1016/j.nedt.2008.10.005.

[5] E. C. Ball, “Annotation an effective device for student feedback: A critical review of the literature,” *Nurse Education in Practice*, vol. 10, no. 3, pp. 138–143, May 2010, doi: 10.1016/j.nepr.2009.05.003.

[6] R. H. Palmer, M. K. Moulton, R. H. Stone, D. L. Lavender, M. Fulford, and B. B. Phillips, “The impact of synchronous hybrid instruction on students’ engagement in a pharmacotherapy course,” *Pharmacy Practice*, vol. 20, no. 1, pp. 1–8, Jan. 2022, doi: 10.18549/PharmPract.2022.1.2611.

[7] F. J. Aiken, “The benefits and challenges of using blended learning and hybrid learning from different perspectives,” *Open Learning*, vol. 40, no. 3, pp. 231–233, Jul. 2025, doi: 10.1080/02680513.2025.2512464.

[8] J. Chen, P. Lai, A. Chan, V. Man, and C.-H. Chan, “AI-Assisted Enhancement of Student Presentation Skills: Challenges and Opportunities,” *Sustainability*, vol. 15, no. 1, p. 196, Jan. 2023, doi: 10.3390/su15010196.

[9] J.-O. Yoon and M. Kim, “The Effects of Captions on Deaf Students’ Content Comprehension, Cognitive Load, and Motivation in Online Learning,” *American Annals of the Deaf*, vol. 156, no. 3, pp. 283–289, 2011, doi: 10.1353/aad.2011.0026.

[10] A. Shimada, F. Okubo, C. Yin, and H. Ogata, “Automatic Summarization of Lecture Slides for Enhanced Student Preview—Technical Report and User Study,” *IEEE Transactions on Learning Technologies*, vol. 11, no. 2, p. 165, Apr. 2018, doi: 10.1109/TLT.2017.2682086.

[11] Microsoft, “Present from PowerPoint Live in Microsoft Teams,” Microsoft Support. [Online]. Available: https://support.microsoft.com/en-us/office/present-from-powerpoint-live-in-microsoft-teams-28b20e74-7165-499c-9bd4-0ad975d448ad. [Accessed: Apr. 9, 2026].

[12] Zoom, “Free Online Whiteboard: Collaborate in Real Time,” Zoom. [Online]. Available: https://www.zoom.com/en/products/online-whiteboard/. [Accessed: Apr. 9, 2026].

[13] H. Alhusban, “A Novel Synchronous Hybrid Learning Method: Voices from Saudi Arabia,” *Electronic Journal of e-Learning*, vol. 20, no. 4, pp. 400–418, Oct. 2022, doi: 10.34190/ejel.20.4.2340.

[14] S. S. Kumari, B. W. D. Raja, and S. R. Sundaravalli, “Analysis Paralysis – The Product of Information Explosion,” *Annals of the Romanian Society for Cell Biology*, vol. 25, no. 4, pp. 4456–4458, Jan. 2021.

[15] G. Lee and X. Zhai, “Realizing Visual Question Answering for Education: GPT-4V as a Multimodal AI,” *TechTrends*, vol. 69, no. 2, pp. 271–287, Mar. 2025, doi: 10.1007/s11528-024-01035-z.

[16] Microsoft, “Visual Studio Tools for Office (VSTO),” Visual Studio Documentation, 2024. [Online]. Available: https://learn.microsoft.com/en-us/visualstudio/vsto/. [Accessed: Dec. 30, 2025].

[17] Microsoft, “Office Add-ins platform overview,” Office Dev Center, 2024. [Online]. Available: https://learn.microsoft.com/en-us/office/dev/add-ins/overview/office-add-ins. [Accessed: Dec. 30, 2025].

[18] D. T. Campbell, J. C. Stanley, and N. L. Gage, *Experimental and quasi-experimental designs for research*. Chicago: R. McNally, 1966.

[19] T. D. Cook and D. T. Campbell, *Quasi-experimentation : design & analysis issues for field settings*. Boston: Houghton Mifflin, 1979.

Appendix A. User Study Questionnaire

Instructions to participants:

Please complete the questionnaire after finishing the assigned scenario. For Part B, rate each statement on a 1–5 scale, where higher scores indicate stronger agreement or higher perceived usefulness.

Scenario codes

A = Basic Live Lecture

B = Unlock + Ink

C = Replay + Summary

Response legend for Part A

Completed successfully

Completed with issue

Not completed / not applicable

Participant details

Field	Response
Participant ID	_____
Role	Instructor / Student
Date	_____
Scenario completed	A Basic Live Lecture / B Unlock + Ink / C Replay + Summary
Device used	Laptop / Desktop / Tablet
Browser	_____

Part A

Part A1. Join and setup

No.	Item	Response
A1	I could access the viewer or presenter flow without major difficulty.	Completed successfully / Completed with issue / Not completed / not applicable
A2	The session started correctly after setup.	Completed successfully / Completed with issue / Not completed / not applicable
A3	The viewer loaded the correct presentation title and first slide.	Completed successfully / Completed with issue / Not completed / not applicable

A4	The session status was clear to me (waiting / live / ended).	Completed successfully / Completed with issue / Not completed / not applicable

Part A2. Live session behaviour

No.	Item	Response
A5	Slide changes appeared correctly during the live session.	Completed successfully / Completed with issue / Not completed / not applicable
A6	The current slide indicator or navigation state was clear.	Completed successfully / Completed with issue / Not completed / not applicable
A7	Locked slides were visibly different from unlocked slides.	Completed successfully / Completed with issue / Not completed / not applicable
A8	I understood when a slide became newly available.	Completed successfully / Completed with issue / Not completed / not applicable
A9	Ink annotations appeared in the correct place on the slide.	Completed successfully / Completed with issue / Not completed / not applicable
A10	Animation-related content appeared correctly during the session.	Completed successfully / Completed with issue / Not completed / not applicable
A11	Live transcript/captions were visible when speech was present.	Completed successfully / Completed with issue / Not completed / not applicable
A12	The viewer layout remained clear while multiple features were active.	Completed successfully / Completed with issue / Not completed / not applicable

Part A3. Replay and artifacts

No.	Item	Response
A13	I could open the replay successfully after the session ended.	Completed successfully / Completed with issue / Not completed / not applicable
A14	Replay content matched what I saw in the live session.	Completed successfully / Completed with issue / Not completed / not applicable
A15	Ink appeared correctly in replay.	Completed successfully / Completed with issue / Not completed / not applicable
A16	Transcript was available in replay.	Completed successfully / Completed with issue / Not completed / not applicable
A17	AI summary was available or clearly marked unavailable.	Completed successfully / Completed with issue / Not completed / not applicable
A18	I could find the download area easily.	Completed successfully / Completed with issue / Not completed / not applicable
A19	I could download the annotated PDF, transcript, or other artifacts successfully.	Completed successfully / Completed with issue / Not completed / not applicable

Part A4. Overall task completion

No.	Item	Response
A20	I could complete the assigned scenario without external intervention.	Completed successfully / Completed with issue / Not completed / not applicable
A21	I would be able to repeat this task in a real class situation.	Completed successfully / Completed with issue / Not completed / not applicable

Part B

Part B instructions

Rate each statement from 1 to 5.
Use Not applicable / unavailable only where stated.

Part B1. Baseline pain points (Students)

No.	Item	Scale
B1	Slides are often unavailable to me during the live lecture.	1 / 2 / 3 / 4 / 5
B2	I often lose access to instructor annotations after class.	1 / 2 / 3 / 4 / 5
B3	It is difficult to review lectures because transcript, replay, or annotated materials are missing.	1 / 2 / 3 / 4 / 5
B4	I often need multiple tools to follow hybrid lectures.	1 / 2 / 3 / 4 / 5
B5	I feel less engaged as a remote student than in-person students.	1 / 2 / 3 / 4 / 5

Part B2. Baseline pain points (Instructors)

No.	Item	Scale
B6	Sharing slides during or after class is time-consuming.	1 / 2 / 3 / 4 / 5
B7	I lack good control over when students can see each slide.	1 / 2 / 3 / 4 / 5
B8	My annotations are usually lost after teaching.	1 / 2 / 3 / 4 / 5
B9	I need multiple disconnected tools to run a hybrid class.	1 / 2 / 3 / 4 / 5
B10	Supporting remote and in-person students equally is difficult.	1 / 2 / 3 / 4 / 5

Part B3. Usability

No.	Item	Scale
B11	The system was easy to start using.	1 / 2 / 3 / 4 / 5

B12	The viewer interface was intuitive.	1 / 2 / 3 / 4 / 5
B13	The interface was visually clear and well organised.	1 / 2 / 3 / 4 / 5
B14	I could understand what each major feature did.	1 / 2 / 3 / 4 / 5
B15	I encountered no major confusion during the session.	1 / 2 / 3 / 4 / 5

Part B4. Live features

No.	Item	Scale
B16	Slide changes felt prompt.	1 / 2 / 3 / 4 / 5
B17	Ink annotations appeared quickly and accurately.	1 / 2 / 3 / 4 / 5
B18	Progressive slide unlocking was easy to understand.	1 / 2 / 3 / 4 / 5
B19	Live captions/transcript were useful.	1 / 2 / 3 / 4 / 5
B20	The layout remained usable during live teaching.	1 / 2 / 3 / 4 / 5

Part B5. Replay and artifacts

No.	Item	Scale
B21	Replay reflected the live session accurately.	1 / 2 / 3 / 4 / 5
B22	Replay preserved ink and animations correctly.	1 / 2 / 3 / 4 / 5
B23	The downloadable artifacts were easy to find.	1 / 2 / 3 / 4 / 5
B24	The downloaded files were useful.	1 / 2 / 3 / 4 / 5
B25	The AI summary was useful.	1 / 2 / 3 / 4 / 5 / Not applicable / unavailable

Part B6. Pain alleviation and productivity

No.	Item	Scale
B26	AutoSlide reduces the problem of slides being unavailable during class.	1 / 2 / 3 / 4 / 5
B27	AutoSlide reduces the problem of lost annotations.	1 / 2 / 3 / 4 / 5
B28	AutoSlide improves post-class review through replay, transcript, and artifacts.	1 / 2 / 3 / 4 / 5

B29	AutoSlide reduces the need to use multiple separate tools.	1 / 2 / 3 / 4 / 5
B30	AutoSlide improves remote student engagement.	1 / 2 / 3 / 4 / 5
B31	AutoSlide would improve my productivity in a real teaching or learning workflow.	1 / 2 / 3 / 4 / 5

Part B7. Comparative value

No.	Item	Scale
B32	Compared with receiving static slides after class, this system provides more value.	1 / 2 / 3 / 4 / 5
B33	I would recommend this system to others in a hybrid learning context.	1 / 2 / 3 / 4 / 5
B34	I would be willing to use this system again.	1 / 2 / 3 / 4 / 5

Part C

Part C. Short-answer feedback

No.	Question	Response
C1	What was the most useful feature?	
C2	What was the most frustrating or confusing part?	
C3	Did the system reduce any pain point you commonly experience in hybrid teaching or learning? Which one?	
C4	What still remains unsolved?	
C5	What one improvement would most increase your productivity?	
C6	Any additional comments?	

Observer notes

Observer / researcher record

No.	Item	Response
O1	User hesitation observed	Yes / No
O2	Needed assistance	Yes / No
O3	Major failure occurred	Yes / No
O4	Notes	Free text